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"It has to be fun"

– Health promotion through reduced sedentary time in older adults transitioning from working life to retirement

Lisa Hultman



Mälardalen University

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– HEALTH PROMOTION THROUGH REDUCED SEDENTARY TIME IN OLDER ADULTS TRANSITIONING FROM WORKING LIFE TO RETIREMENT

Lisa Hultman

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Department of Health Sciences, Innovation and Design

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Akademisk avhandling

som för avläggande av filosofie doktorsexamen i fysioterapi vid Institutionen för hälsovetenskap, innovation och design kommer att offentligen försvaras onsdagen den 10 juni 2026, 10.15 i Gamma, Mälardalens universitet, Västerås.

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Abstract

Globally, society is undergoing a demographic shift characterised by an ageing population and increased life expectancy, posing challenges to the healthcare system that may be addressed through health promotion interventions supporting beneficial health behaviours. The transition from working life to retirement could be a potential window for such interventions. Sedentary behaviour, common among older adults, is associated with adverse health outcomes. Although eHealth interventions incorporating self-management strategies may support reductions in sedentary time, there is a lack of research targeting reduced sedentary time during the transition to retirement. Involving end users in the development of eHealth interventions through participatory design may help ensure alignment with users' needs. The overall aim was to contribute to health promotion by generating new knowledge on how sedentary time can be reduced, and how adherence to reduced sedentary time can be promoted, through an eHealth intervention including self-management strategies, in older adults transitioning from working life to retirement. A further aim was to develop and test such an intervention.

The thesis comprises four studies following a participatory design approach. In the initial needs-assessment phase, focus groups with older adults transitioning to retirement explored perceptions of self-management strategies for reducing sedentary time (Study I). In the idea generation phase, workshops were conducted to explore desired features of an eHealth intervention including self-management for reducing sedentary time and promoting adherence to the reduced sedentary time among the target group and to develop a digital prototype of such an intervention (Study II). In the testing and retesting phase, the feasibility and usability of the developed eHealth intervention were examined through a mixed-methods evaluation study which applied an ABA single-case experimental design (Study III). In the evaluation phase the effect of the eHealth intervention on daily sedentary time, assessed using an accelerometer, was examined through a randomised controlled trial (RCT) (Study IV).

The findings highlight the importance of supporting autonomy in eHealth interventions for reducing sedentary time through adaptable self-management strategies and user-directed intervention use (Studies I–II). Reducing sedentary time required support for making changes in everyday life. Self-management strategies addressing both affective and cognitive determinants – especially identifying joyful activities – were identified as central in reducing sedentary time and supporting adherence (Studies I–II). The GAUdiS (an acronym for the Swedish equivalent of Joyful Activities Facilitate Reduced Sedentary Time) eHealth intervention was developed, comprising features for identifying joyful activities and supporting goalsetting and planning based on mental contrasting with implementation intentions, which can be understood based on the Integrated Behavior Change model (Study II). Although GAUdiS was developed together with older adults transitioning to retirement as active participants, it demonstrated only modest feasibility and usability, possibly partly due to the demands associated with a standalone eHealth design (Study III). The final RCT showed no reduction in sedentary time with GAUdiS among older adults transitioning to retirement; these findings should be interpreted in light of the small sample size (Study IV).

In conclusion, the findings suggest that employing autonomy-supportive approaches, as well as targeting both affective and cognitive determinants, are considered central to reducing sedentary time in the transition to retirement. However, for an eHealth intervention to be effective in reducing sedentary time in the transition for retirement, additional components beyond those offered by the current standalone version of GAUdiS may be needed. Given that the studies included in the thesis involved a limited number of participants, further research including larger groups is needed in order to examine how sedentary time can be reduced during the transition to retirement. While this thesis focused on total sedentary time, future research might explore whether GAUdiS could be effective when targeting only those sedentary behaviours that individuals perceive as unwanted or unhealthy, in alignment with health promotion values.

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“It does not do to dwell on dreams and forget to live, remember that.”
Albus Dumbledore
— J. K. Rowling*, Harry Potter and the Philosopher’s Stone*

List of Papers

This thesis is based on the following papers, which are referred to in the text by their Roman numerals.

I Hultman, L., Eklund, C., von Heideken Wågert, P., Söderlund, A., Elfström, ML. (2024) It Has To Be My Way-Reducing Sedentary Time in the Transition to Retirement. *J Aging Phys Act*, 20:1–10.

II Hultman, L., Eklund, C., von Heideken Wågert, P., Söderlund, A., Lindén, M., Elfström, ML. (2025) Development of an eHealth Intervention Including Self-Management for Reducing Sedentary Time in the Transition to Retirement: Participatory Design Study. *JMIR Form Res,* 20:9:e63567.

III Hultman, L., Eklund, C., von Heideken Wågert, P., Söderlund, A., Elfström, ML. (2026) One felt like a newly hatched bird - Feasibility and Usability of an eHealth Intervention Including Joyful Activities for Reducing Sedentary Time in the Transition to Retirement: Mixed-Method Single-Case study. *Submitted.*

IV Hultman, L., Eklund, C., von Heideken Wågert, P., Söderlund, A., Elfström, ML. The effect of an eHealth web-based Intervention for Reducing Sedentary Time in Older Adults in Transition to Retirement – Randomized Controlled Trial. *In manuscript.*

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Abbreviations

BBQ	Brunnsviken Brief Quality of life
EQ VAS	EuroQol visual analogue scale
EQ-5D-5L	EuroQol five-dimension Questionnaire
ESES-S	Swedish Exercise Self-Efficacy Scale
FRT	Fantasy realization theory
GP-Core	General Population Clinical Outcomes in Routine Evaluation-Outcome Measure
HADS	Hospital Anxiety and Depression Scale
IBC model	Integrated behavior change model
IPAQ	International Physical Activity Questionnaire
LASA	Longitudinal Aging Study Amsterdam sedentary behavior
LMM	Linear mixed models
MAUQ	mHealth App Usability Questionnaire
MC	Mental contrasting
MCII	Mental contrasting with implementation intention
NCD	Non-communicable diseases
OATRs	Older adults transitioning to retirement
PD	Participatory design
PDR	Participatory design research
PR	Participatory research
RCT	Randomised controlled trial
S-MAUQ	Swedish mHealth App Usability Questionnaire
SCED	Single-case experimental design
SDT	Self-determination theory
ST	Sedentary time
TPB	Theory of planned behaviour
TA	Thematic analysis
WHO	World Health Organization
WWT	Wake wear time

Introduction

Doing research can be described as going on an adventure. I am thrilled that you have chosen to put time in reading the story of this research adventure. The adventure takes place in a world where previous research and the medical and technical development make us live longer than previous generations. Which of course is great news! We get to spend more time on the journey of life, and we have the older adults in our lives for a longer time of that journey. Today do most people in the western society have 20-30 years of life ahead to look forward to after time of retirement. It is thereby important to make healthy life choices to maintain health and function for as long as possible.

Research and the technical development have changed work, transport and how spare time is spent. Many work tasks are less mobile and more sedentary, it is less common to walk or bike to or from work, and the time spent in front of screens has increased in recent years. Overall, more time is spent in sedentary behaviour which research has shown is not in line with healthy life choices. Instead, to spend a lot of time in sedentary behaviour increases the risk of several non-communicable diseases and premature death.

So how can this development be changed? How can sedentary time be reduced to increase health in older adults? Times for major changes in life such as retirement can be an advantageous time to reduce sedentary time as daily life changes. So how can sedentary time be reduced when newly retired? Let's find out! Let the adventure begin.

Background

Health promotion from the perspective of physiotherapy with a behavioural medicine approach

Health promotion is defined by the World Health Organization (WHO) as "the process of enabling people to increase control over, and to improve their health" (World Health Organization, 2021, p. 4). This makes it important to describe what health is. This thesis adopts a holistic view and the biopsychosocial model of health whereby biological, psychological and social aspects are considered equally important in the assessment of health (Engel, 1977). It agrees with the WHO's definition of health as "a state of complete physical, mental and social wellbeing and not merely the absence of disease or infirmity" (World Health Organization, 1948, p. 1). In this, health is not viewed as an elusive utopia; instead, this thesis agrees with the assumptions of person-centred care (Ekman, 2014), in which a person's own perception of their health is a crucial point of view when deciding when and whether health is achieved. This places the power of defining what health is and phrasing health-related goals in the hands of the individual.

There is an increasing proportion of older adults in the global population. (World Health Organization, 2015). This demographic shift means upcoming challenges to health and welfare systems, with a smaller proportion of the population being of working age and thus able to contribute financially to or be employed within the welfare system. At the same time, non-communicable diseases (NCDs) such as cancer, cardiovascular disease, chronic respiratory disease and diabetes are increasing and are the most common cause of ill health and death globally (NCD Countdown 2030 collaborators, 2018). This is an additional aspect of the importance of health promotion as the desired outcome of improved health is not only beneficial at an individual level but also at a societal level. To alleviate the challenges of the demographic shift, it is important that the older population stay as healthy as possible; this can be achieved through health promotion interventions to support individuals in their health behaviours.

Increased life expectancy also implies that a larger proportion of one's life is spent in retirement. Earlier research has found that many health behaviours change in retirement (Barnett et al., 2012; Ding et al., 2016).

Smeaton et al. (2017) found that older adults in their study both saw their retirement as an opportunity for improved health behaviours but also as a potential risk of negative health behaviours. Research on older adults in Sweden found that retirement is a major life change which reshapes opportunities and motivation and causes changes to habits in older adults (Karlsson, Lundell, et al., 2025). Retirement has previously been proposed to be a beneficial time for health promotion interventions in older adults (Vigezzi et al., 2021). Taken together, these findings suggest that the transition to retirement may represent an opportune period for introducing health promotion interventions.

Health promotion is an important and necessary part of the Swedish healthcare system, and the National Board of Health and Welfare state that all welfare professions within the healthcare system, when suitable, should provide the patient with information regarding the prevention of unhealthy health behaviours (The National Board of Health and Welfare, 2024) – health behaviour being a person's behaviour with the goal of promoting, protecting or regaining health, regardless of whether the behaviour is objectively healthy (World Health Organization, 2021).

The physiotherapist plays an important role in the health-promotive work within health and welfare, having the ability to provide both knowledge and skills regarding the promotion of several health behaviours (Bezner, 2015). The aim of physiotherapy is to promote health and reduce illness and suffering by helping individuals maintain or regain optimal movement, function and participation in society (Fysioterapeuterna, 2019). A person-centred approach is a central part of the physiotherapy process (Fysioterapeuterna, 2019). Health promotion and the person-centred approach share the understanding that the individual's resources, abilities and health behaviours are important assets for achieving health, indicating a need for a deeper understanding of health behaviour. This need can be met by adopting a behavioural medicine approach to physiotherapy. Behavioural medicine is a multidisciplinary field involving the integration of behavioural and biomedical knowledge that is relevant to health and disease, and can be applied for prevention, health promotion, diagnosis, treatment, rehabilitation, and care. Behaviour knowledge refers to the knowledge of how health- and disease-related behaviours are affected by societal, psychosocial, existential, environmental, economic and cultural processes. Biomedical knowledge refers to processes related to pathological, physiological and medical processes (Dekker et al., 2021). Health promotion interventions using a behavioural medicine approach have shown to be promising in changing health behaviour in relation to reducing the risk of NCDs, such as reducing sedentary time (ST) and increasing physical activity (Hannan et al., 2021).

As mentioned earlier, health promotion is a matter of enabling and empowering individuals to take control over and improve their health, making self-management important. Self-management is a person's own day-to-day management of health promotion activities or chronic disease, and involves the management of medicines, emotions, meaningful behaviours and roles (Lorig & Holman, 2003). Health promotion interventions have proven to be more effective when they include self-management strategies (Barlow et al., 2002).

In health promotion interventions, adherence has been addressed as a matter of concern (World Health Organization, 2003). Adherence refers to the degree to which an individual's health behaviours correspond to the recommendations they have agreed upon together with a healthcare provider (World Health Organization, 2003, p. 3). Different concepts have been used to understand and describe adherence. Historically, *compliance* has been used, but this term has been criticised for giving the individual a more passive role as a recipient of recommendations and treatments (Bissonnette, 2008). In contrast, *adherence* offers a more inclusive approach whereby the individual and the caregiver together agree on the goals and interventions; this better reflects the values of patient-centred care. There has been a semantic shift from "do they do as we tell them to" to a more inclusive, person-centred approach: "do they choose to do as we agreed upon". A less frequently used concept related to *adherence* is that of *concordance*, which, even more than adherence, highlights the individual's role in decision-making: the power differences between caregiver and individual are noted, talking about a partnership in decision-making and enlightened communication (Snowden et al., 2014). However, researchers have found that the three concepts are used synonymously and that there is inconsistency in how they are described (Snowden et al., 2014). Being a more widespread concept, *adherence* is used in this thesis, although the importance of partnership and shared power from the concept of *concordance* are included in the term.

In summary, health promotion interventions for improving health in older adults are important. When designing such interventions, a behavioural medicine approach can be used to understand the complexity found in health behaviour. These interventions should target older adults transitioning to retirement (OATRs) and include elements of self-management. Moreover, promoting adherence to the desired health behaviour is essential for achieving meaningful behaviour change. However, health behaviour is a broad concept, raising the question of which specific behaviour should be addressed.

Sedentary behaviour

The health behaviour of interest in this thesis is sedentary behaviour, which can be defined as "any waking behaviour with an energy expenditure <1.5 metabolic equivalents (METs), while in a sitting, lying or reclining position" (Tremblay et al., 2017). ST is all the time one spends in sedentary behaviour (Tremblay et al., 2017). Sedentary behaviour is a separate behaviour from physical inactivity, although it is not seldom described synonymously with it in the literature. Physical inactivity instead entails not reaching the recommended 150 to 300 minutes of moderate aerobic physical activity or 75 to 150 minutes of vigorous aerobic physical activity per week (Bull et al., 2020). Sedentary behaviour is a natural part of our everyday life, but spending an extended amount of time being sedentary increases the risk of NCDs and premature death (Ajufo et al., 2025; Biswas et al., 2015; Ekelund et al., 2019; Rossen et al., 2020). A specific level for when ST is considered harmful has not been defined. However, research has found a dose-response association with an increased harmful effect at 9.5 hours of daily ST (Ekelund et al., 2019) while other researchers have found an increased risk at 6-8 hours (Patterson et al., 2018). The harmful effects of extended ST can be alleviated by performing regular physical activity (Ekelund et al., 2016), and recent research has found that reducing ST with 30 minutes can lower the risks for those who are also physically active at a low level (Ekelund et al., 2026). Older adults (>60 years) in industrialised countries spend an average of 9.4 hours a day in sedentary behaviour (Harvey et al., 2015). According to statistics based on self-reported data from the public health agency in Sweden, 24% of older adults aged 65–84 in Sweden are sedentary for 7-9 hours a day and 14% are sedentary for 10 or more hours (Public Health Agency of Sweden, 2024).

Not only does ST increase the risk of disease and premature death; Wilson et al. (2019) found that older adults who spent less time in objectively measured ST reported higher self-rated health. There is conflicting evidence regarding the association of ST with mental health. While Sansano-Nadal et al. (2021) found no association between ST and self-reported mental health, other researchers have found that older adults with high levels of ST reported lower mental health than those who spent less time sitting did (Bampton et al., 2015). Further, Jiang et al. (2024) found that total ST was associated with depression, but found differences when ST was separated into mentally active and mentally inactive ST, with mentally active ST – such as reading, socialising while seated or playing games – being associated with decreased depression (Jiang et al., 2024).

Although time in sedentary behaviour has been found to increase with age (Harvey et al., 2015) until recently fairly little was known about sedentary behaviour in the transition to retirement (Sprod et al., 2015). Recent research

suggests that ST increases during this transition. (Leskinen et al., 2018). When using objectively measured ST Suorsa et al. (2020) found that women increased their ST after retirement and that women from manual occupations increased their ST the most. Men's ST did not change after retirement but were found to remain at high levels after retirement (Suorsa et al., 2020).

Given that older adults and OATRs commonly spend a considerable amount of time being sedentary, and considering the negative consequences associated with sedentary behaviour, health promotion interventions targeting a reduction of ST among OATRs are required. In order to reduce ST among OATRs, knowledge is required regarding their perceptions of sedentary behaviour, their views on interventions aimed at reducing ST, and what strategies have previously been used to address ST in this population. Let us now explore what is currently known.

Older adults' perception of sedentary behaviour

Older adults are often unaware of how much time they spend in sedentary behaviour, and of the difference between sedentary behaviour and physical inactivity (Compernolle et al., 2020; McGowan et al., 2021). An interview study with community-dwelling older adults found that those who considered themselves to be more active spoke judgmentally regarding those who spend more time in sedentary behaviour, and that participants in the study described feelings of guilt when they had spent a great deal of time in sedentary behaviour (McGowan et al., 2021). Research has found that older adults say that doing things they enjoy increases their ST (Compernolle et al., 2020) while other researchers have found positive affect, such as enjoyment, to be a facilitator for performing non-sedentary behaviours (McGowan et al., 2019). That affect also influences sedentary behaviour in other populations, is supported by the work of Phipps et al. (2022), whose findings suggest that sedentary behaviour is a largely affectively driven behaviour among workers.

Research exploring what older adults do while they are sedentary found that older adults described high-value purposeful sedentary behaviour when socialising, resting or being cognitively active, in contrast to low-value passive sedentary (Palmer et al., 2019). This aligns with recent research showing that OATRs may experience sedentary behaviour as both positive and healthy when it reflects their own choice and provides an opportunity to rest, but also as negative and unhealthy when it is associated with loneliness or boredom (Eklund et al., 2021).

Although recent years have seen some research on how older adults experience sedentary behaviour, a knowledge gap remains regarding OATRs' perceptions of self-management strategies for reducing ST.

Interventions for reducing sedentary time

Sedentary behaviour is distinguished from physical inactivity not only due to the differences in definition themselves but also due to behavioural change components of the two distinct behaviours. When trying to change physical inactivity in health promotion interventions, the goal is to increase a behaviour (moderate to vigorous physical activity). Another aim is for the person to perform the behaviour for a limited time span (for example 20–60 minutes) for a couple of days during the week, or daily. When the aim is to change sedentary behaviour, on the other hand, the goal is to reduce a behaviour. Furthermore, rather than happening only on limited occasions during the week, the desired behaviour change extends over all waking hours of the day. Due to these differences, interventions for reducing ST are likely to differ from those aiming to increase physical activity. Earlier research has also found that interventions for reducing ST are more effective if they target ST specifically rather than increased physical activity (Prince et al., 2014).

Reducing ST is a matter of changing a habitual behaviour (Maher & Dunton, 2019). A habitual behaviour is a behaviour which is controlled by habit, in which *habit* can be described as a stimuli or cue creating an automatic impulse to perform the behaviour (Gardner, 2015). With this in mind, reducing ST is not only a matter of breaking the behaviour in the moment but also of disrupting the connection between cue and behaviour, making *habit disruption* more suitable wording than *breaking a habit*, as habit disruption indicates a need for changes in executive processes.

Habit researchers discuss four different approaches for achieving habit disruption: inhibition, which involves wilfully, through self-regulation, choosing not to perform the behaviour; habit discontinuation, meaning avoiding the cues; habit degradation, which is a matter of weakening the cue-response association; and habit substitution, in which the unwanted behaviour is replaced with a more desired alternative (Gardner et al., 2023). Other literature providing guidance in how to choose strategies for habit disruption proposes avoiding cues (habit discontinuity) when possible and, when not possible, inhibiting unwanted response (Habit inhibition) or pursuing alternative behaviour (habit substitution) (Gardner et al., 2020). Recommended self-management strategies for disrupting habits include planning inhibitory responses, distraction, monitoring of behaviour, and making environmental changes in order to avoid cues (Gardner et al., 2023). These strategies have also been used frequently in interventions for reducing ST in older adults (Petrusevski et al., 2020).

A systematic review of interventions for reducing ST in community-dwelling older adults found that ST might be reduced by those targeting sedentary behaviour, but the evidence remains uncertain (Chastin et al., 2021). In their scoping review, Petrusevski et al. (2020) found that multi-

level interventions can reduce ST in community-dwelling older adults. Their research shows goalsetting, information and self-monitoring to be the most common behaviour change techniques used, and information and action planning were used in the most successful intervention (Petrusevski et al., 2020).

ST has been conceptualised not only as a behaviour distinct from physical activity and as a habitual behaviour, but also as one that is influenced by affective processes. The question, then, is to what extent interventions targeting affect have been employed to reduce ST.

Affect and interventions for reducing sedentary time

Although sedentary behaviour has been characterised as an affectively driven behaviour, research examining how affect can be targeted to reduce ST remains scarce. Lithopoulos et al. (2020) compared the effect of information regarding affective outcomes, information regarding instrumental outcomes, self-regulation and a control group, and found that participants in the affective group reduced their ST the most in the short term but that there was no difference between the groups at the 3-month follow-up. However, in recent years, interventions targeting affective determinants for other health behaviours have been developed and tested in several studies (Forster et al., 2018; Rhodes et al., 2019; Ruissen et al., 2019; Ruissen et al., 2018).

Social cognitive theories are widely applied in research as theoretical bases for understanding behaviour, but have been criticised for not describing how affect influences our behaviour (Ferrer & Mendes, 2018). Health behaviour is influenced by affective determinants which can be divided into four categories (Stevens et al., 2020; Williams & Evans, 2014): incidental affect, affective response, affect processing and affectively charged motivation. Incidental affect is emotional state that is not related to the health behaviour. Affective response includes both immediate as well as a post-behavioural affective response to a behaviour. Affect processing can be both automatic, through for example implicit affective attitudes, and reflective, through for example anticipated affect. Affectively charged motivation is influenced by, for example earlier affective responses and processing hedonic motivation and implicit motivation (Stevens et al., 2020).

Although interventions addressing affective determinants of health behaviours have shown promising results, there are no studies specifically targeting affective determinants in order to reduce ST among OATRs.

Welfare technology and eHealth

Welfare technology, developed to meet the upcoming challenges within the welfare system, aims to contribute to higher quality and effectiveness and can be defined as any digital technology which aims to maintain or increase activity, participation or independence in a person who has or is at risk of impairment (The National Board of Health and Welfare, 2015). The digitalisation of healthcare entails using technology as a means to provide healthcare without compromising quality, accessibility, efficiency or equity (Odone et al., 2019). An example of this digitalisation is the use of eHealth. eHealth includes all health services and health-related information that is delivered through the internet or internet-related technologies (Eysenbach, 2001). eHealth interventions are a broad concept including but not limited to mobile device-delivered messages, and the use of stationary computers or phone calls, and have previously been used for health promotion (De Santis et al., 2023; Romeo et al., 2019) reducing ST (Dunn et al., 2018) and providing self-management strategies (Chatterjee et al., 2021; Geuens et al., 2019). Researchers have found that older adults' adherence to health promotion interventions is higher when delivered through eHealth. This might be explained by high levels of enjoyment when using eHealth interventions (Valenzuela et al., 2018).

Yerrakalva et al. (2019) reported potential benefits of eHealth in reducing ST in older adults but acknowledged that the findings were uncertain. A recent systematic review examining the effectiveness of eHealth interventions delivered from mobile devices on ST in older adults found that such eHealth can be used to reduce ST in that population, and that the included studies used the latest technologies, including smartphones, tablets and smart watches (Chen et al., 2025). A systematic review of eHealth interventions, including health coaching targeting OATRs found that the literature was limited and that few eHealth interventions specifically targeted OATRs. They described the area as a research gap, and recommended that OATRs be allowed to include their perspective in the development phase (Stara et al., 2020). When developing new eHealth interventions, the requirements for quality, accessibility, efficiency, and equity must be achieved. This could be done by using an approach which involves the end users in the design process. By doing so there is a better chance of designing useful eHealth in line with the needs and requests of the end users (Clemensen et al., 2017).

Participatory design as a framework

Participatory research (PR), a research approach with the aim of taking action, has its roots in action research. PR includes designs, methods and frameworks which strive to include those who are affected by the research, who have real-life experience of the research question and are the focus of the research. Central in PR is its view on end users or individuals affected by the research as active participants rather than subjects, and the approach that research is performed "with" them rather than "on" them (Vaughn & Jacquez, 2020). Although it is connected to action research and the work of Kurt Lewin (1890-1947) (Reason & Bradbury, 2008), PR was further developed in Scandinavia during the 1970s through Kristen Nygaard's work on computer development and use in the industrial sector, which brought together industrial workers, trade-union representatives and researchers (Binder et al., 2009).

One of the PR approaches is participatory design research (PDR), a user-centred approach with an iterative design process in which products or services are designed in collaboration with the end users (Vaughn & Jacquez, 2020). Participatory and user-centred approaches have been widely used in designing eHealth (Moore et al., 2019) and is widely used in designing eHealth for older adults, including the development of exercise programmes, as well as fall, rehab and prevention interventions and communication interventions (Merkel & Kucharski, 2019).

Participatory Design (PD) is an approach within the broader field of PDR. This project uses PD as described by Clemensen et al. (2017) as a framework whereby representatives of the end users, web designers and researchers are equal participants in the development of eHealth. PD shares values with action research in regard to its views on knowledge and participation as well as the distribution of power. The researchers' and web developers' academic knowledge, and the experience-based knowledge of the end users are valued equally. End users, web developers and researchers are all equivalent participants in the research, and are all referred to as participants. Power is recognised and concrete action is taken in the process in order to create a balance of power and to share power between participants (Clemensen et al., 2017).

PD consists of four phases; the first, need assessment, aims to identify the participants' needs; the second, idea generation, is the phase in which ideas and prototypes are generated through collaboration, with everyone's knowledge and ideas regarded as equally valuable. These phases are followed by the testing and retesting phase, which entails testing and further development, and the evaluation phase, in which the intervention's effectiveness is tested (Clemensen et al., 2017).

Within PD, the intention is to foster active participation whereby participants feel free to share their experiences and knowledge, which can be supported by agreeing on frameworks, rules, a shared language and goals (Denecke et al., 2025). Various tools and methods have been developed for this purpose (Vandekerckhove et al., 2020).

A systematic review of involving older adults in technology design found that this group's involvement can increase knowledge and enable adaptation of the technology during the design process, as well as give the older adults participating in the study a feeling of worth and participation. Further, the review's results showed that older adults' participation might lead to increased acceptability, but that this was not the case in all studies included (Fischer et al., 2020).

Models and theories for behaviour change

Integrated behavior change model

eHealth interventions are often based on a model or theory of behaviour change. Commonly used theories are social cognitive theories (Ajzen, 1991; Bandura, 1986). These theories have received criticism concerning the intention-behaviour gap, in regard to the assumption that the intention to perform a behaviour is sufficient to perform the behaviour (Rhodes & de Bruijn, 2013). Instead, an integrated theory for health-behaviour change can be used, combining elements from developed theories to fill the gaps found by earlier research. Such a theory is the Integrated Behavior Change (IBC) model (Hagger & Chatzisarantis, 2014).

The IBC model includes functions of deliberative, conscious processes, volitional processes of planning, and nonconscious, implicit processes. The foundation of the model is the Theory of Planned Behaviour (TPB). Intention is a key construct within the TPB, and is made up of the belief-based components of attitudes, beliefs regarding social norms and perceived behavioural control (Hagger & Chatzisarantis, 2014). Perceived behavioural control includes the two underlying concepts of self-efficacy and controllability (Ajzen, 2002). The IBC model explains the origin of the belief-based components as levels of autonomous motivation. Autonomous motivation, originating from self-determination theory (SDT), entails motivation that arises from experiencing that one is acting in line with one's true self and by one's own choice, and that one's behaviour fulfils basic psychological needs (Deci & Ryan, 1985, 2000; Hagger & Chatzisarantis, 2014).

To fill the intention-behaviour gap, the model adds a volitional process of action planning following intention. An action plan is formed by finding

a cue in the environment that is planned to work as an offset of a desired behaviour. Lastly, nonconscious, implicit processes are added to the model by including implicit attitudes and implicit motivation. These influence behaviours that do not require planning or consideration, such as when the behaviour is easy to perform or is a habitual action. Implicit motivation is closely linked to autonomous motivation, influencing our behaviour not only through cognitive but also implicit processes (Hagger & Chatzisarantis, 2014).

Affect plays a natural role within the IBC model due to the concept of autonomous motivation from SDT. A concept connected to autonomous motivation is intrinsic motivation, meaning that a person performs a behaviour due to their satisfaction with and enjoyment of the behaviour itself. Autonomous motivation within the IBC model is thus, through intrinsic motivation, associated with the affective determinations of affective response, affect processing and affectively charged motivation (Stevens et al., 2020). Also, positive affect related to affective response is a key outcome within SDT, as those who are autonomously motivated tend to experience satisfaction and positive affect when performing the desired behaviour (Deci & Ryan, 1985). Ryan and Deci (2001) also describe how autonomous motivation is closely linked to wellbeing. Eudaimonic wellbeing comes from personal growth, self-realisation and the experience that one's actions are in line with one's values, while hedonic wellbeing entails happiness and positive emotions being the goal.

Affective determinants within the affect processing category can further be included in the IBC model as attitudes within the IBC model could include both cognitive and affective attitudes. Furthermore, the implicit processes included in the IBC model include affect as implicit motives refer to an implicit process related to autonomous motivation) and implicit attitudes, which can include implicit affective attitudes.

Fantasy realisation theory and mental contrasting with implementation intention

In the past there has been an understanding that positive thinking leads to goal fulfilment and behaviour change, but this has been countered by other research which states that fantasising about the future does not lead to behaviour change (Oettingen, 2012; Oettingen et al., 2001). This was the starting point for the development of Fantasy Realisation Theory (FRT). FRT distinguishes between two ways we think about the future: through expectations or through fantasies. Expectations are judgments of how likely it is that an event or behaviour will take place, often based on previous experience. Fantasies are free thoughts regarding the future, regardless of the

credibility of the occurrence of the imagined event or behaviour. Researchers have found that merely fantasising about a desired future does not lead to behaviour change. While fantasising or thinking positively about the future can be a strategy for calming down in a stressful situation, it does not lead to either happiness or goal achievement in the long run. Within FRT, goalsetting is viewed as a means of self-regulation. Goals can be set through three different routes: the first is to simply fantasise about the desired future; the second is to merely dwell upon the downside of the present reality; and the third, which is the only one that leads to high goal achievement, includes elements of expectations. In this route, called mental contrasting (MC), a fantasy or desired future is contrasted with the present reality. This results in the need to act in order to take the leap from the present to the desired future, which requires an assessment of one's expectations regarding being able to reach the goal. High expectations will lead to a high goal commitment, and low expectations will lead to a low goal commitment (Oettingen, 2012; Oettingen et al., 2001). When conducting MC, cognitive or affective, outcomes and obstacles can be considered. Research has found that considering affect may further enhance goal commitment and goal achievement in MC (Ruissen et al., 2019; Ruissen et al., 2018). The effect of MC can be further increased by adding an element of planning, which fills the intention-behaviour gap. The implementation intentions adds a plan for how to overcome the obstacles in order to fulfil the goal by adding an "if…-then" plan. This is called Mental Contrasting with Implementation Intentions (MCII) (Oettingen, 2012). MCII increases goal fulfilment through contrasting the desired future with the present reality, and also offers support in goal formulation and action planning. The person or patient is asked to think of a wish regarding the future, then to think about the best possible outcome if this wish is fulfilled. The next step is to think about likely obstacles that can arise and hinder the person in fulfilling the goal. The implementation intention is then added by asking the person to make a plan for how to overcome or avoid this obstacle (Oettingen, 2012). A systematic review has found that MCII may serve as a brief intervention for improving health behaviours (Cross & Sheffield, 2019).

Rationale

Health promotion is a matter of enabling people to increase control over, and to improve their health, where health is not merely the absence of disease, but ratter concerns a person's own assessment of their physical, psychological, and mental well-being. The welfare system faces several challenges due to a growing older population. One way to address these challenges is to support older adults in engaging in advantageous health behaviours through health promotion interventions to improve their health. One health behaviour that has received increasing attention in recent years is sedentary behaviour.

Research has found that to spend a lot of time in sedentary behaviour is common among older adults. Sedentary behaviours are all behaviours performed while in a seated, lying or reclining position while having a low energy expenditure. Spending a lot of time of the day in sedentary behaviours is associated with poorer health and an increased risk of premature death. Health promotion interventions supporting older adults to reduce the amount of time they spend being sedentary would thereby be beneficial both on an individual and a societal level.

The transition to retirement is a period during which health behaviours might change. This transition may therefore be a favourable time to introduce interventions for reducing ST among older adults.

Earlier research has found that ST might be reduced using eHealth interventions, ideally incorporating elements of self-management and focusing on adherence to support lasting behaviour change. Currently, there is a knowledge gap regarding how ST can be reduced and adherence to reduced ST can be supported using eHealth interventions for older adults in transition to retirement. Further no eHealth intervention has been specifically designed for this purpose.

When designing new eHealth interventions there is a need to involve end-user as equal participants in the design process in order to develop an intervention that is in line with their needs. By using participatory design, new knowledge can be created, new eHealth interventions developed, and evaluated, and the needs of older adults in transition to retirement can be included in the design process.

Aims

The overall aim was to contribute to health promotion by generating new knowledge on how sedentary time can be reduced, and how adherence to reduced sedentary time can be promoted, through an eHealth intervention including self-management strategies, in older adults transitioning from working life to retirement. A further aim was to develop and test such an intervention.

The specific aims of each of the included studies were:

Study I — To explore perceptions among older adults transitioning from working life to retirement regarding self-management strategies for reducing sedentary time and adhering to the reduced sedentary time.

Study II — To explore desired features of an eHealth intervention including self-management for reducing sedentary time and promoting adherence to reduced sedentary time in older adults transitioning from working life to retirement. Further, the aim was to develop a digital prototype of such an eHealth intervention.

Study III — To examine the feasibility and usability of an eHealth intervention including self-management for reducing sedentary time and promoting adherence to reduced sedentary time in older adults transitioning from working life to retirement.

Study IV — To examine the effect of an eHealth intervention including self-management for reducing sedentary time and promoting adherence to reduced sedentary time in older adults transitioning from working life to retirement.

Methods

This thesis uses participatory design (PD) as its methodological framework and includes a combination of qualitative, quantitative and mixed methods. PD includes four phases: need assessment, idea generation, testing and re-testing, and evaluation (Clemensen et al., 2017). For an overview of how the included studies relate to the PD framework, see Table 1.

Table 1. Overview of study design, data collection, and analysis methods of the studies as well as how they relate to the phases of PD.

Study	Design	Data collection	Data analysis	Phase of PD
I	Qualitative explorative	Focus group interviews	Qualitative content analysis	Need assessment
II	Qualitative explorative	Workshops	Thematic analysis	Idea generation
III	Mixed method Experimental Single-case	Accelerometer measurements, questionnaires, researcher fieldnotes	Descriptive, visual, and statistical analysis, thematic analysis	Testing and retesting
IV	Experimental comparative	Accelerometer measurements, questionnaires	Descriptive and linear mixed models	Evaluation

Participants and settings

OATRs participated in all phases of the project. Also, two researchers and one web developer participated in Study II. OATRs were recruited from retirement organisations' meetings and theme days for retirees, from an interview in a local newspaper, through information on municipalities' digital intranets for employees, by advertising on social media and by sending information about the study to retirement organisations and companies with the request to spread it to their members and employees. Also, participants were encouraged to inform other potential participants about the study. The two participating researchers were involved in the project as researchers, and the web designer worked at a company that had contacted the university with an interest in collaborating.

The criteria for inclusion of OATRs were: age 60-75 years, able to speak and understand Swedish, and able to read and comprehend the study instructions. For participants who were still working, an added inclusion criterion was that they were planning to retire within 3 years, and those who were already retired had to have been retired for a maximum of 5 years. The criteria for exclusion were: presence of a serious disease, severe loss of vision, and severe loss of communicative ability. For the last two studies, an additional inclusion criterion was ≥6 hours of daily ST, and the exclusion criteria were contraindications to increased physical activity, not being able to walk 500 meters without support and having used a self-management intervention targeting ST or physical activity within the past year. Participants received written and verbal information regarding the study and written consent was signed with pen and paper in all studies.

A total of 81 participants were included in the project. Participating OATRs' demographics can be found in Table 2. A participant flowchart from Study IV can be found in Figure 1.

Table 2. Demographics of participating OATRs.

Demographics		Study I *n*= 28	Study II *n*=6	Study III *n*= 5	Study IV *n*=39	
					Intervention group *n*= 20	Control group *n*= 19
Age, mean (SD); min-max		66.9 (2.9); 62-74	67.2 (1.9); 64-70	64.6 (0.89); 64-66	67.95 (4.42); 62-76	65.37 (3.29) 61-73
Sex, *n* (%)						
	Female	21 (75)	4 (67)	4 (80)	16 (80)	11 (61)
	Male	7 (25)	2 (33)	1 (20)	4 (20)	7 (39)
Living conditions, *n* (%)						
	Cohabiting	21 (75)	3 (50)	3 (60)	13 (68)	15 (79)
	Living alone	7 (25)	3 (50)	2 (40)	6 (32)	4 (21)
Type of housing, *n* (%)						
	House/Townhouse	-	3 (50)	3 (60)	13 (65)	10 (59)
	Apartment	-	3 (50)	2 (40)	7 (35)	7 (41)
Employment, *n* (%)						
	Retired	23 (82)	6 (100)	4 (80)	8 (40)	5 (28)
	Working	5 (18)	0 (0)	1 (20)	9 (45)	10 (56)
	Part time retired	-	-	-	3 (15)	3 (17)
Education, n (%)						
	Elementary school	-	-	0 (0)	0 (0)	0 (0)
	High school	-	-	2 (40)	7 (37)	8 (44)
	University	-	-	3 (60)	12 (63)	10 (56)
Most sedentary or nonsedentary during the workday at current or last employment, *n* (%)						
	Sedentary	11 (39)	3 (50)	5 (100)	17 (85)	16 (84)
	Nonsedentary	17 (61)	3 (50)	0 (0)	3 (15)	3 (16)
Sedentary time, mean (SD); min-max						
	Sedentary time during weekdays,[a]	9.4 (4.9); 3.8–20.5	-	-	-	-
	Sedentary time during weekends,[a]	9 .5 (3.4); 4–19	-	-	-	-
	Self-reported total daily ST	-	5.75 (2); 2-8	-	-	-
Long-term health problems, *n* (%)						
	Yes	-	-	1 (20)	7 (37)	7 (37)
	No	-	-	4 (80)	12 (63)	12 (63)
Fall during the last 12 months, *n* (%)						
	Yes, once	-	-	2 (40)	1 (5)	2 (11)
	No	-	-	3 (60)	19 (95)	17 (89)

[a]Self-report of ST using the Longitudinal Aging Study Amsterdam Sedentary Behavior (LASA) Questionnaire (Visser & Koster, 2013)

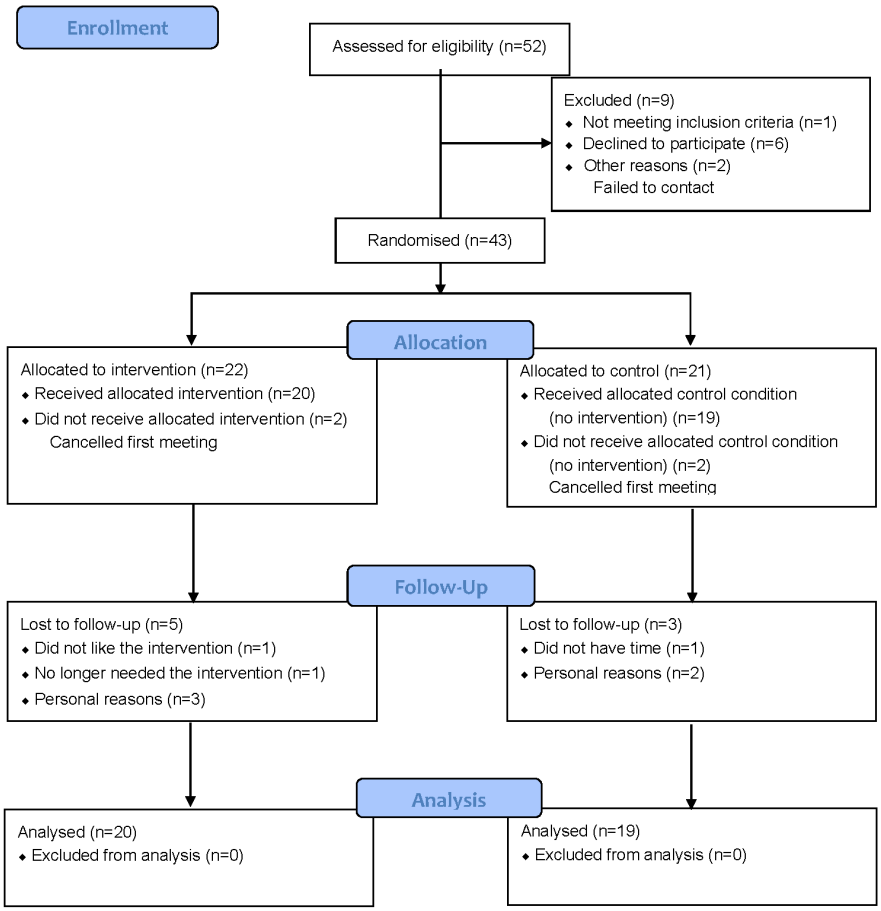


Figure 1. Participant flowchart from Study IV. Follow-up refers to follow-up 3-month post intervention introduction.

Need assessment (Study I)

Data collection

To explore the perception among OATRs self-management strategies for reducing ST and increasing adherence to reduced ST, four focus group interviews were conducted (Krueger, 2015). The groups consisted of seven to nine participants each. A semi-structured interview guide was designed, consisting of questions within the following areas: how to reduce ST; how to advise someone else who wants to reduce their ST; how to adhere to

reduced ST; and the acceptability of delivering self-management strategies for reducing ST through a digital solution. The interview guide was designed by the researchers and was tested in two pilot group interviews (n=2+2), and no changes were made to the guide after the pilot testing. The focus group interviews took place at the university during October and November 2019, and lasted 87-106 minutes (mean=97, SD=8). All interviews were audio-recorded, and an assistant took notes during the interviews. The participants' demographic data was collected using a study-specific questionnaire at the beginning of each focus group interview.

Data analysis

Data from the focus group interviews was analysed using latent qualitative content analysis (Graneheim & Lundman, 2004), using NVivo QSR International (2018). Audio recordings were transcribed verbatim. Transcribed interviews and notes taken from the focus group interviews were read through several times to get an overall understanding of the content. Texts that addressed the aim of the study were marked as meaning units, condensed and abstracted into codes. Codes were then grouped into subcategories and categories based on commonalities. Lastly, an overarching theme for all categories and subcategories was described. This was carried out in an iterative process among all researchers in the study.

Idea generation (Study II)

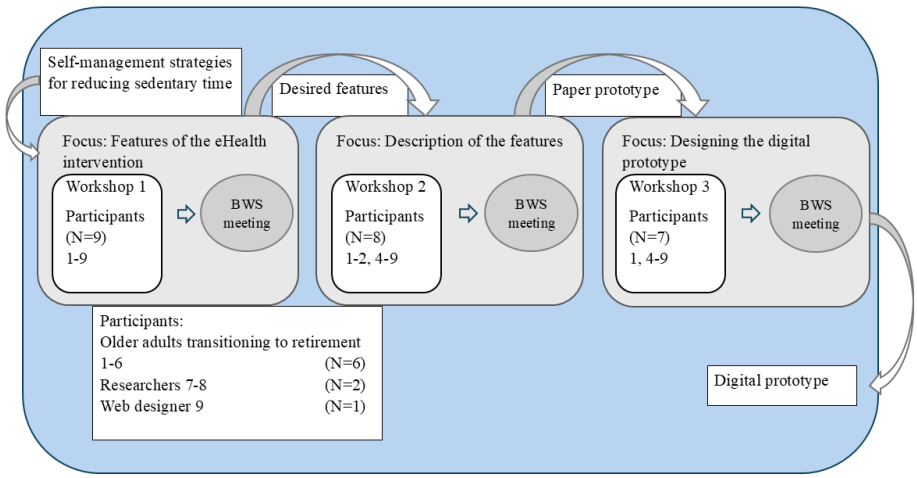
Data collection

To explore desired features and develop a digital prototype of an eHealth intervention including self-management for reducing ST and promoting adherence to reduced ST in OATRs, three workshops were conducted between November 2021 and January 2022. An experienced workshop moderator led the workshops to improve the balance of power between the participants. The workshops were conducted in a meeting room at the university, and were designed by the researchers participating in the workshops and the moderator.

Participants' demographic data was collected using a questionnaire at the beginning of the first workshop. The findings from Study I, including self-management strategies for reducing ST in the transition to retirement, were used as a basis for planning the first workshop. The subsequent workshops were based on the preliminary findings from the previous workshop. At each workshop, established workshop tools were used (Vandekerckhove et al.,

2020). At the first workshop the PD tool Cards was used, with participants being asked to list the features they would desire in a digital solution for operationalising self-management strategies for reducing ST. Workshop 2 used the PD tool 2D Mapping, with participants working with a mind map of features. A paper prototype was used as a starting point at the beginning of Workshop 3, and was further developed using the PD tool User Journey. Participants were also asked to prioritise the importance of the features discussed by selecting the first, second and third most important features for reducing ST. After this prioritisation, drafts of each feature were presented and suggested changes to the paper prototype were discussed. The workshops were audio-recorded, and photos were taken of the mind maps and Post-it notes created during the workshops.

After each workshop, 'between-workshop meetings' were conducted with the two researchers participating in the workshops as well as four additional researchers with extensive experience in the development of eHealth, behaviour change interventions, and geriatric research. Preliminary findings and proposed solutions were discussed at these meetings, and after the last workshop the theoretical framework of the eHealth intervention was discussed. For a visual overview of the workshop process, see Figure 2.



BWS meetings=between-workshops meetings.

Figure 2. Workshop process map from Study II.

Data analysis

The audio recordings from the three workshops were transcribed verbatim. The data analysis was conducted in two steps using inductive thematic analysis (TA) (Braun & Clarke, 2021) using NVivo (Lumivero, 2023). In the first step, data from the first and second workshops was included (feature analysis), with a focus on features discussed and descriptions of these features. This analysis was used to verify that nothing had been missed or misunderstood in the paper prototype which had been presented at Workshop 3. Data from Workshop 3 was analysed separately in the change analysis, which focused on changes that needed to be made to the paper prototype as well as the final layout. Both analyses followed the six phases of TA (Braun & Clarke, 2021). As part of the first three phases, audio recordings and transcriptions were listened to and read through several times to get familiar with the data. Relevant segments were coded at a semantic level. Codes were then grouped together in initial themes, and the writing began. Initial themes were checked against the whole data set and discussed among all researchers in the study (Phase 4 of TA). The themes were named and the writing of the themes developed into the writing of the manuscript, as in Phases 5 and 6 of TA.

Testing and retesting (Study III)

Data collection

In the testing and retesting phase of the project, the eHealth intervention's feasibility and usability were tested through a mixed-methods evaluation study (Creswell & Inoue, 2025), including ABA single-case experimental design (SCED) (Krasny-Pacini & Evans, 2018). The study took place during February and March 2023, and lasted four weeks. Five participants used the intervention in order to evaluate its feasibility and usability. In this project, feasibility is understood as the intertwined result of acceptability, suitability and participant response (Orsmond & Cohn, 2015). Data regarding acceptability, referring to the users' thoughts and feelings when using the eHealth intervention (Perski & Short, 2021) and the included self-management strategies (Yardley et al., 2015), were collected using two questionnaires completed by the participants. The first questionnaire was completed following the participants' use of the intervention and was referred to as an acceptability diary. The second questionnaire was completed after participants had used the eHealth intervention for 2weeks and was referred to as the acceptability questionnaire. When completing the acceptability diary, participants

were also asked to make an acceptability rating from 1 to 10 concerning whether it was a positive experience to use the intervention on that occasion (1=Do not agree at all and 10=Fully agree). The researcher also kept field notes of issues involving how participants described their use and experience of the intervention, questions asked by the participants, and participants' reactions to using the intervention. Suitability was evaluated by observing user logins to the intervention during the intervention phase. Participant response to using the intervention is a combination of qualitative feedback from the user and the preliminary effect of the target behaviour (Orsmond & Cohn, 2015); in this case, changes in ST. To evaluate the preliminary effect, ST was assessed using an Axivity AX3 accelerometer. Accelerometer measurements were conducted in line with the ProPASS guidelines (ProPASS consortium, 2020). The accelerometer was placed on the participant's thigh for the 7-day baseline phase, during the 2-week intervention phase, and during the 7-day withdrawal phase. The accelerometers were replaced with a fully charged device once a week, and in-between the participants were asked to remove it themselves once and place it on the opposite thigh to minimise the risk of skin irritation. Participants were asked to log what time they went to bed at night and what time they got out of bed in the morning. The ActiPASS software (Hettiarachchi & Johansson, 2025) was used to process data from the accelerometers. In ActiPASS the validated algorithm Acti4 is used to identify ST by evaluating the accelerometer's movements and inclination in relation to the line of gravity (Skotte et al., 2014; Stemland et al., 2015). Non-wear and sleep were identified by the ActiPASS software (Johansson et al., 2023). The sleep algorithm was applied to diary-defined in-bed periods. To be included in the analysis, a day must include a minimum of 20 hours of wear time, at least one walking period, and a sleep interval (Blodgett et al., 2024).

Usability is defined as 'the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use' (ISO9241-11, 2018). Usability was assessed using the Swedish version of the mHealth app usability questionnaire (MAUQ) (Zhou et al., 2019). There are four versions of the MAUQ targeting standalone or interactive apps and for healthcare providers or patients; here, the version for standalone apps for patients was used. The MAUQ is a validated and reliable tool for assessing usability, and consists of 18 items with a seven-level answer scale from *Strongly disagree* to *Strongly agree* (Figure 7) (Zhou et al., 2019). The Swedish version of the MAUQ (S-MAUQ) (Karlsson, Sönnerfors, et al., 2025), available for interactive apps. One item was translated and added to this version to be consistent with the MAUQ targeting standalone apps for patients.

Participants received access to the eHealth intervention at the second meeting with the first author of the study, held on the first day of the intervention phase. Each participant met with the first author of the study separately. Participants were asked to log in by themselves, and if they needed assistance this was recorded in the researchers' field notes. Participants were then asked to use the intervention ad libitum, meaning according to their own preferences, during the 2-week intervention phase. For an overview of the procedure followed in Study III, see Figure 3.

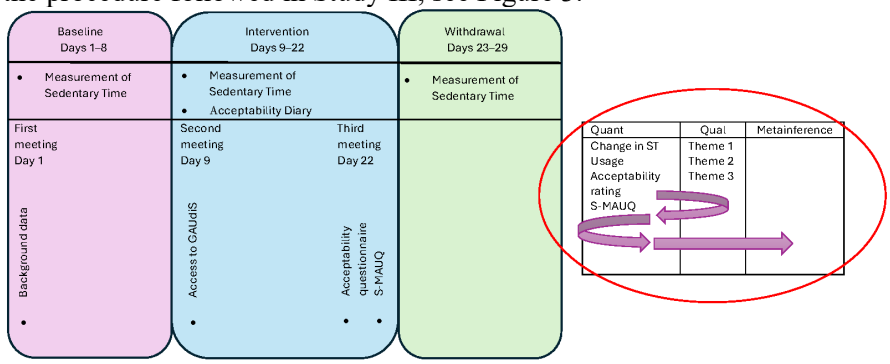


Figure 3. Study procedure including a joint display of the data analysis. The red circle highlights the integration of the two methodologies.

Data analysis

Qualitative analysis

Analysis of the qualitative data was performed using inductive TA (Braun & Clarke, 2021) using Nvivo (Lumivero, 2023). The six phases of TA were followed using the same procedure as in Study II. The unit of analysis was data from the acceptability questionnaire, acceptability diaries and researchers' field notes.

Quantitative analysis

Data on participant response concerning change in ST was analysed visually and statistically. To compensate for differences in wake time and sleep between days, the daily percentage of wake-wear time (WWT) spent in ST was calculated and visualised in a graph for each participant. Included in the graph was data regarding the participant's use and acceptability rating of the eHealth intervention. Data was visually analysed regarding level, trend and variability within and between each phase (Kratochwill et al., 2010). Data was also analysed with Tau-U (Parker et al., 2011) using R version 4.3.0 (2023-04-21 ucrt) packages Readxl, Ggplot2 and Openxlsx. Tau-U is used to statistically analyse effect size by calculating the overlap between two

phases. Tau-U is suitable when there is a high variability of data, and can correct for trend within the baseline phase (Parker et al., 2011). The S-MAUQ was calculated for each participant separately, and a mean was also calculated for all participants. Data was visualised separately for each item and for each participant in order to obtain a more in-depth understanding of the eHealth intervention's usability.

Metainference

Qualitative and quantitative analyses were joined in a specific, elaborative metainference (Younas et al., 2025), with the first author of the study going back and forth between the different analyses to gain an understanding of the intervention's feasibility and usability. This process of methodological integration is presented in the joint display shown in Figure 3.

Evaluation (Study IV)

Data collection

In the evaluation phase, a randomised controlled trial (RCT) was conducted to examine the effect of the developed eHealth intervention.

Included participants were randomly assigned to the intervention or control group (1:1) using computer-generated permuted block randomisation (Urbaniak & Plous, 2013). Participants who lived together were assigned to the same group.

The primary outcome measure was changes in ST, measured using an AX3 accelerometer for 7 full days following the same procedure for measurement and processing of data as in Study III. Secondary outcome measures were self assessment of ST; self-efficacy and motivation for, and importance of reducing ST; ability to perform physical activities; exercise self-efficacy; self-rated health; health-related quality of life; subjective well-being, anxiety, and depression; and physical activity. Further information regarding outcome measures can be found in Table 3. Use of the intervention was analysed as active days, referring to the number of days during which a participant used any of the intervention's features. Participants were included in the study from August to December 2023. Data was collected at baseline and at follow-up at 3 months.

Table 3. Primary and secondary outcome measures in Study IV, names of instrument, number of items, minimum and maximum values, guidance on interpreting the scores, and Cronbach's alpha.

Outcome	Measurement	Scoring and interpretation	Self-report	Cronbach's alpha
Primary Outcome				
Sedentary time	Accelerometer AX3	Daily total ST as percentage of wake wear time.		NA
Secondary Outcome				
Self-reported sedentary time	Longitudinal Aging Study Amsterdam sedentary behavior questionnaire, LASA (Visser & Koster, 2013)	Mean average ST in hours per day.	X	NA
Physical activity	International physical activity questionnaire, IPAQ (Craig et al., 2003)	MET minutes per week. Higher values indicate higher physical activity.	X	NA
Self-rated health	EQ VAS from EuroQol five-dimension questionnaire, EQ-5D-5L (Herdman et al., 2011)	Higher scores indicate higher self-rated health. Single item question rated 0–100	X	NA
Perceived health-related quality of life	Brunnsviken Brief Quality of life, BBQ (Lindner et al., 2016)	Higher scores indicate greater quality of life. 12 items rated 0–4, summed to total 0–96	X	0.91
Subjective well-being	General Population Clinical Outcomes in Routine Evaluation-Outcome Measure, GP-CORE (Evans et al., 2005)	Lower scores indicate higher well-being. 14 items rated 0–4, giving a mean ranging 0–4	X	0.70
Anxiety and depression	Hospital Anxiety and Depression Scale, HADS (Zigmond & Snaith, 1983)	Scores for anxiety and depression are reported separately. Higher scores indicate higher levels of self-reported anxiety or depression symptoms. 7 items rated 0–3, summed to total 0–21	X	0.70[a] 0.53[d]
Ability for physical activities	Study-specific questions	Higher scores indicate higher physical ability. 6 items rated 1–8, summed to total 6–48	X	0.79
Exercise Self-efficacy	Swedish Exercise Self-Efficacy, ESES-S (Nessen et al., 2015; Resnick & Jenkins, 2000)	Higher scores indicate higher self-efficacy. 6 items rated 1–10, summed to total 6–60	X	0.92
Self-efficacy, motivation for & importance of reducing sedentary time	Study-specific questions	Higher scores indicate higher self-efficacy, motivation and importance. Single item questions rated 1–10	X	NA

[a]HADS-Anxiety; [d]HADS-Depression

Data was collected at individual study visits apart from participants who lived together and chose to be scheduled together. Background and secondary outcome data was collected using a digital questionnaire which participants answered on a provided computer during the study visits. EQ VAS from EQ-5D-5L was collected using paper and pen. After baseline data was collected, including the 7-day accelerometer measurement, participants returned the AX3 and were informed of their group allocation from randomisation. Participants in the intervention group gained access to the eHealth intervention, and were encouraged to view the instruction videos and use the main features once and were then encouraged to use the intervention ad libitum.

Data analysis

Statistical analyses were conducted in R using version 4.3.0 (2023-04-21 ucrt) packages Stringr, Tidyverse, Lubridate, Haven, Nlme, Emmeans, Janitor, Openxlsx, Dplyr, Tibble, lmerTest, ggplot2, ggeffects, lme4, car, Readxl, tableone, and writexl. The intention-to-treat principle was followed, with all randomised participants included in the analyses. Linear mixed models (LMM) were used to account for both fixed effects (e.g. test group, time) and random effects (e.g. subject specific-intercepts), allowing for correlation between repeated measurements within individuals (baseline and one follow-up). Differences between groups at baseline were adjusted for in each model. Internal consistency reliability was examined for each multi-item instrument using Cronbach's alpha, presented in Table 3. Drop-out analyses were performed by comparing completers and non-completers regarding baseline measures for primary and secondary outcomes. Bonferroni-adjusted *p*-values were calculated to control for Type I error (Bland & Altman, 1995).

Ethical considerations

Careful consideration was taken regarding the risk of participants being exposed to physical or mental health risks in any of the included studies. No obvious risk was detected. The risk that participants would experience any discomfort or an invasion of their privacy by participating in discussions in focus groups or workshops was considered low and to be outweighed by the benefits of performing the study. Also, at the beginning of the focus groups and workshops, rules regarding the conversation climate were highlighted.

In Studies III and IV, any activity performed by the participants as a means to reduce ST was selected by the participants themselves. They were advised to always base their decisions on their own abilities, interests and circumstances, and to never deviate from any recommendations from healthcare professionals. The risk that participants in Studies III and IV would experience skin irritation was counteracted by the use of hypoallergenic plasters. Participants also received the recommendation to remove the accelerometer and plaster after 3 to 4 days and place it on the opposite leg, and to remove it at once if they experienced any skin irritation. All data collected in the study was handled with confidentiality. Only one researcher had access to the user data connected to the eHealth intervention and all data was de-identified, with names replaced with codes in Studies III and IV. Participants were informed that participation was voluntary and could be terminated at any time. Verbal and written information regarding the study was given to the participants, and written consent was collected. No participant can be identified from the findings of any of the included studies. All studies were approved by the Regional Ethical Review Board in Uppsala, Sweden, Dnr 2019-03836. An amended application regarding a time extension of the project was approved (Dnr 2024-04204-02). This extension was due to parental leave and delays related to the COVID pandemic.

Findings

Personalisation, daily support and enjoyment as key components of an eHealth intervention (Study I & II)

In the first two phases of this PD project it was found that participants did not want to feel controlled but instead wanted self-management strategies, and an eHealth intervention delivering them, which they could use in their own way rather than being told what to do. The findings revealed a need for an intervention targeting both affective and cognitive determinants of health behaviour, as positive affect was found to be a reason for participating in non-sedentary activities. A person's emotional state could also impact their ambition and ability to try new self-management strategies. Participants described a need for an eHealth intervention which was easy to use and adapted to users with low digital literacy. Accessibility could be improved by including instructions, preferably delivered in writing, audio and video, as everyone takes in information in different ways. It emerged at the workshops that participants wanted an intervention to be accessible from both phones and tablets as well as standard computers in order to increase accessibility, as low digital literacy was considered to be common among OATRs; although high literacy was viewed as common among those who would be retiring in the near future. Studies I and II resulted in the eHealth intervention GAUdiS, which is an acronym of the Swedish for 'joyful activities facilitate reduced ST' (**G**lädjefyllda **A**ktiviteter **U**n**d**erlättar m**i**nskat **S**tillasittande). The word *gaudis* also means "with joy" in Latin. GAUdiS includes self-management strategies, and strives to be adaptable according to the user's joys, interests, preferences and preconditions. GAUdiS is a standalone eHealth intervention, meaning that no additional intervention is provided outside the eHealth intervention.

The first feature for reducing ST in GAUdiS includes information regarding ST. Participants expressed a need for users to receive information regarding sedentary behaviour first, so they would know why it is important to reduce ST. Although it was important for GAUdiS to arouse curiosity and enjoyment in the user at the first impression, it was important that the information be considered trustworthy and come from a reliable source. Information should also be brief, positively formulated and free from blame, and

not cause guilt. This was in reference to the fact that, when using GAUdiS for the first time, the user is presented with a video emphasising the health benefits of reducing ST instead of highlighting the risks associated with prolonged ST, as well as an introduction to GAUdiS.

GAUdiS consists of two overarching features, one of which is a feature for finding joyful activities. Finding joyful activities to do during one's newfound free time as a retiree was considered the most important self-management strategy for reducing ST and adhering to the reduced ST. This was found in both the results of the focus groups and the prioritisation of features in the workshops (Figure 4).

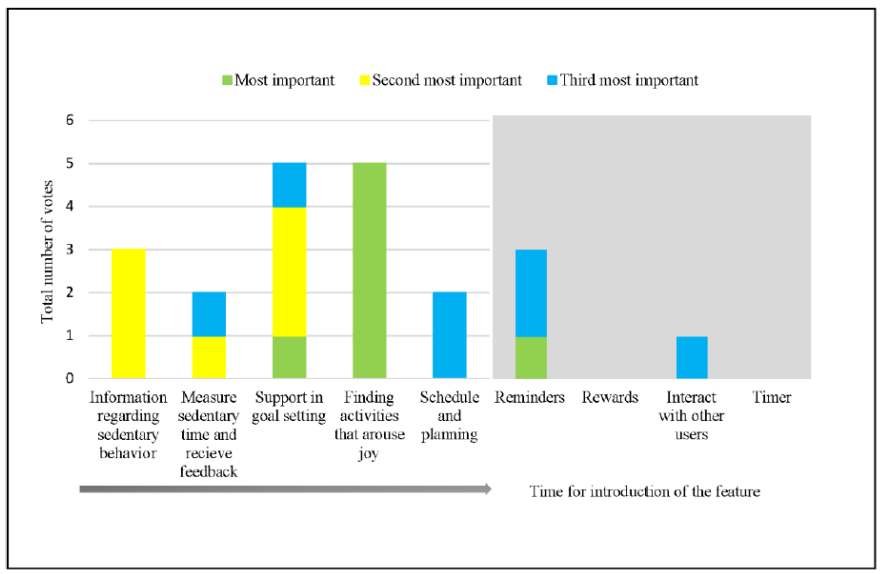


Figure 4. Prioritisation of features and time of introduction and inclusion of features in the paper prototype. The grey area represents functions discussed but not included in the eHealth intervention.

Participants highlighted that what promotes feelings of joy is highly individual, and can include doing activities which are considered fun or rewarding in themselves or otherwise from adding joy-giving components, such as listening to music or an audiobook and doing activities together with others. When using GAUdiS for the first time, the user is asked to select areas of interest, in order to be able to sort and highlight activities that are in line with their interests. Joyful activities are then presented to the user with a short descriptive text and a picture. Participants suggested the possibility of receiving notifications about joyful activities available on a given

day based on their areas of interest, for example, meetings, art exhibitions, or group walks. This would have been challenging to implement due to the substantial time required. Instead, can the user find information and links regarding where activity can be performed in the local area, and is provided with information regarding organisations, interest groups and websites. Participants can save an activity as a favourite with a heart symbol, and add new activities to the eHealth intervention. For an example of the feature for finding joyful activities, see Figure 5.

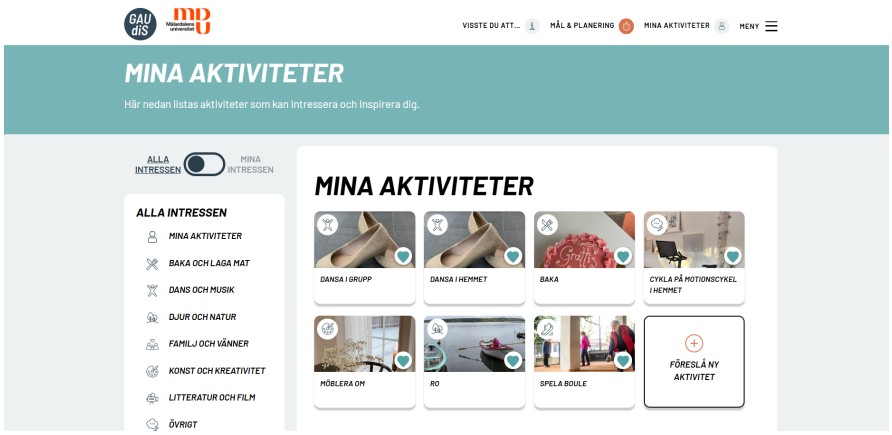


Figure 5. Image from the feature for finding joyful activities in GAUdiS.

Participants in the focus group interviews saw a need to make changes in their everyday life in order to be able to reduce ST in the transition to retirement. It was described as taking the opportunity to be non-sedentary whenever possible and to change the way they perform behaviours which are otherwise done while sedentary. They also expressed a need for support in making these changes. The support could involve setting a goal, making plans and setting a schedule, or making rules for oneself. During the workshops different features for support in reducing ST were discussed; these can be found in Figure 4. The second-most important feature for reducing ST after finding joyful activities was support in goalsetting. Participants expressed that they found it important to set a goal concerning ST but that it was difficult. Prior to setting a goal, participants wanted the eHealth intervention to offer features for measuring ST. This was important, as it would help them judge whether or not it concerned them, and was necessary to be able to formulate the goal. It was also important to them to have the opportunity to measure ST again at follow-up. Different features for measuring ST were discussed. Participants saw advantages to measuring with portable devices such as a smartphone or watch, but found it inconvenient to have to carry a device with them and did not want to include a feature that would entail a

cost to the user. Self-assessing ST on the previous day was considered the most feasible, but should not be too complicated or time-consuming for the user.

These requests are answered by the other overarching feature of GAUdiS, which helps with goalsetting and planning, based on MCII (Oettingen, 2012; Oettingen et al., 2001). Prior to setting a goal, the user assesses their ST using a tool which asks them to estimate how long they were sedentary during set time intervals on the previous day. The user then gets a result displaying their total ST and is prompted to set a goal in four steps. They can choose to listen to or read the instructions for these steps. When setting a goal, the user can choose to set a goal for maximum time to be spent per day in sedentary behaviour for the following day or week. They are then asked to consider how they would feel if they managed to reduce their ST. The next step is to think about obstacles which could prevent them from meeting the goal. The last step is to find solutions for avoiding or overcoming these obstacles. At this point the user can find possible solutions among the tips provided by GAUdiS, or use one of the activities from the feature for finding joyful activities. For an overview of the feature for goalsetting and planning, see Figure 6.

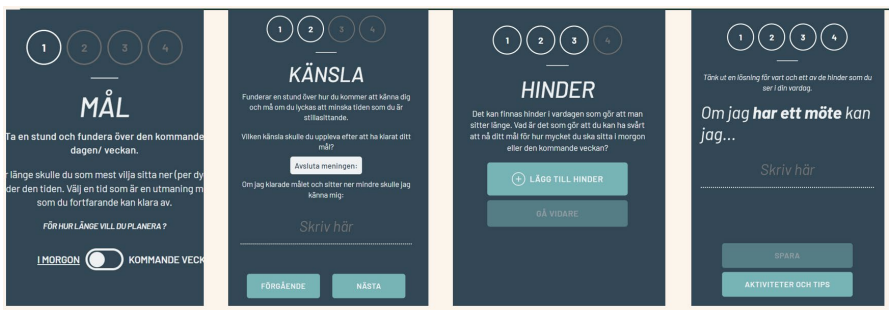


Figure 6. Four images from the feature for goalsetting and planning in GAUdiS.

As participants pointed out the importance of being able to reach GAUdiS from a computer, which led to the choice of using a web app, the opportunity to send out reminders was thought to be lost. Participants expressed disappointment at this, as they wanted to have the opportunity to be reminded regarding the time for goal follow-up and receive a daily reminder about the intervention in general. However, when the digital prototype was being developed into the web app a solution to sensing text messages was found. GAUdiS can send an optional daily text message to the user's phone. These text messages include information on when it is time to follow up on

one's goals, positively formulated information regarding ST, and tips from other OATRs on how to reduce ST which had been collected from the data from, Studies I and II.

Other features which were discussed at the workshops but not included in GAUdiS involved getting rewards upon goal achievement, the opportunity to interact with other users, and the ability to set a timer to limit prolonged ST. Participants discussed the importance of not including too many features, in order to limit the risk of increasing one's screen time rather than reducing ST. They found that the brief feedback they received at goal follow-up was sufficient, and that although they did not interact with others in GAUdiS it contained features for finding activities that would allow them to meet others. A timer was not included in GAUdiS as none of the participants ranked this as an important feature for reducing ST; instead, the information the user receives includes a tip to set a timer when they are sedentary.

Modest feasibility and usability, with participant-guided areas in need of improvement (Study III)

In this study, feasibility was looked at as a combination of the participants' thoughts and feelings in relation to using GAUdiS (acceptability), whether and if so how GAUdiS was used (suitability) and their responses to using GAUdiS, meaning changes in the target behaviour (ST) (Orsmond & Cohn, 2015). The usability of GAUdiS was tested using the S-MAUQ (Karlsson, Sönnerfors, et al., 2025; Zhou et al., 2019).

Four of the included participants participated throughout the four-week study period. One participant (Participant 1) had to end their participation after 2 weeks doe to personal reasons. Data from acceptability diaries acceptability questionnaire and field notes were included in analysis.

The TA regarding the acceptability of GAUdiS resulted in three themes, Inspiration which can lead to change, An interplay with the dynamics of learning, and Frustration when the app doesn't work like I expect it to.

Participants saw potential in GAUdiS. They described the included features as providing awareness and being motivational. Both the feature for finding joyful activities and the one for goalsetting and planning were mentioned as helpful in reducing ST. The acceptability of GAUdiS was shaped by the underlying motivations of the participants, the learning process of getting to know a new eHealth intervention, and individual opportunities. Bugs led to frustration at GAUdiS not working as intended. Participants also expressed feelings of frustration when they wanted GAUdiS to function differently than it was designed to do.

The mean S-MAUQ score was 4.1. Variation was seen in scores on the items. However, some patterns were observed in the S-MAUQ scores: no participants agreed, and three participants disagreed or strongly disagreed, regarding being able to recover easily and quickly after having made a mistake (Q5); no participant agreed that GAUdiS adequately provided knowledge regarding progress of action (Q8); all participants disagreed or answered not applicable regarding being able to use GAUdiS when one's internet connection was poor or not available (Q17); three participants agreed or strongly agreed that it was easy to learn how to use GAUdiS (Q2), that navigation was consistent between screens (Q3) and that the interface allowed them to use all functions (Q4); lastly, one participant strongly disagreed, but three participants agreed or strongly agreed, that GAUdiS would be useful for their health and wellbeing (Q13). For further information regarding the S-MAUQ scorings, see Figure 7.

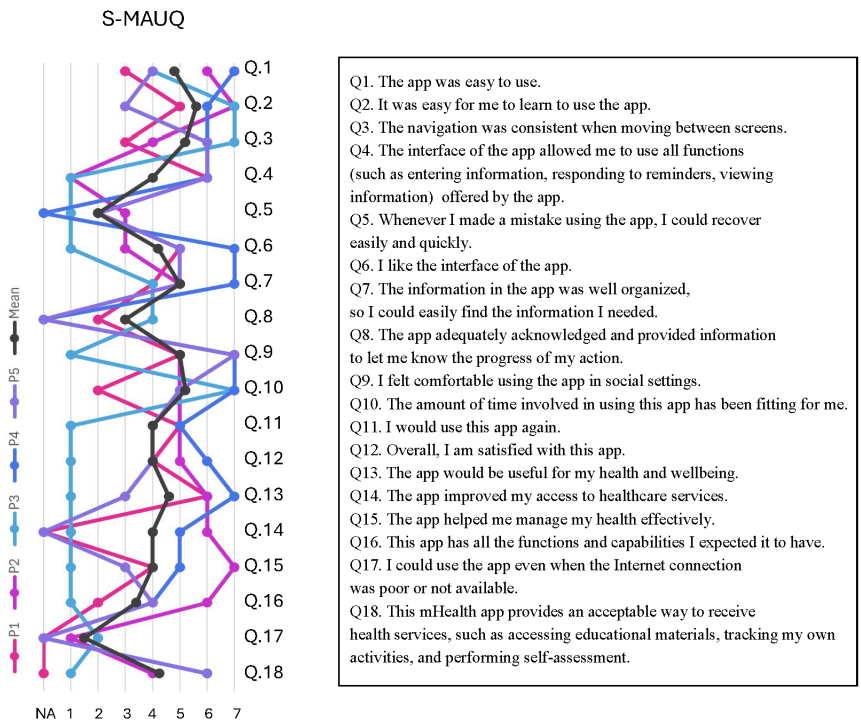


Figure 7. Results from the S-MAUQ, where NA=Not applicable, 1=Strongly disagree, 2=Disagree, 3=Somewhat disagree, 4=Neither agree nor disagree, 5=Somewhat agree, 6=Agree, 7=Strongly agree.

Two participants reported using GAUdiS all but 2 days during the intervention period. Acceptability ratings rarely varied more than 2 points during the intervention period, apart from those of one participant, who rated 5 at Day 1 and 10 at Day 13. For further information regarding acceptability scores, see Figure 8.

Regarding participant response as changes in ST, great variation in daily ST was seen in all phases for all participants, apart from the baseline phase for Participant 4 (P4) (Figure 8). Tau-U showed no significant changes in ST between baseline and intervention or between baseline and withdrawal phase for any of the participants (Table 4).

Participant 2 (P2) showed little difference in level between phases and no trend in baseline or intervention phase, but a negative trend during the withdrawal phase (Figure 8).

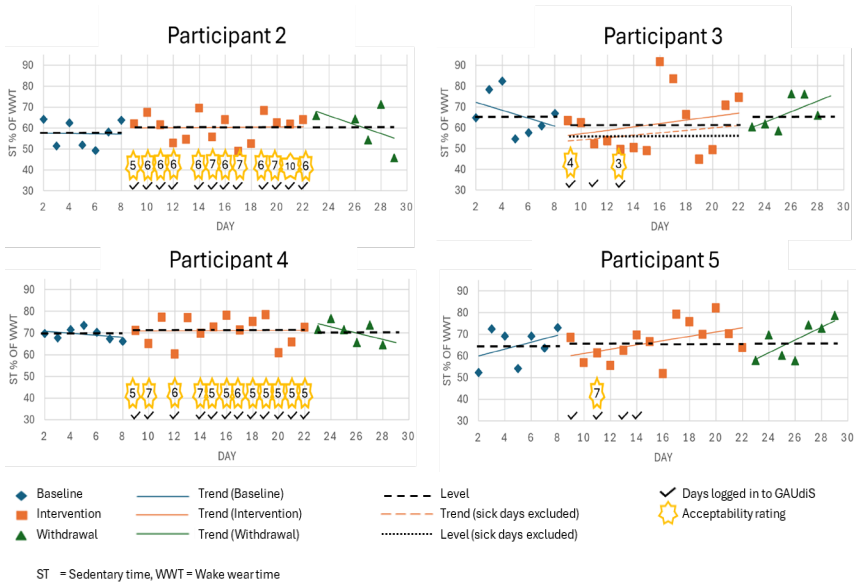


Figure 8. ST measured using Axivity, use and acceptability assessed using acceptability ratings from 1 to 10 each day GAUdiS was used.

Participant 3 (P3) reported sick for 2 days of the intervention phase (Days 16 and 17). Trend and level were visualised with sick days included and excluded. Tau-U was also calculated for both conditions (Table 4). P3 showed a negative trend in baseline followed by a positive trend in both the intervention and withdrawal phases. A slightly lower level of ST was seen in the intervention phase, which was greater with sick days excluded (Figure 8).

The level stayed quite consistent though all phases for P4. A negative trend was seen in the withdrawal phase. For P5 a positive trend was seen in all three phases, but the level did not change between phases (Figure 8).

Table 4. Tau-U analysis for changes between baseline and intervention as well as baseline and withdrawal for all participants.

	BASELINE VS INTERVENTION		BASELINE VS WITHDRAWAL	
	Tau-U	*p*	Ta-U	*p*
P2	0.20	0.65	0.32	0.50
P3A	-0.35	0.41	-0.05	0.95
P3B	-0.57	0.19	-0.05	0.95
P4	-0.09	0.84	-0.10	0.88
P5	0.32	0.46	0.50	0.27

A=sick days included; B=sick days excluded

The metainference suggests that the feasibility and usability of GAUdiS were modest, indicating a need for further development. Although differences were found between participants regarding both the feasibility and usability of GAUdiS, some patterns emerged that indicated a need for further development in three areas: ensuring clarity and ease of use, considering the learning process, and providing supportive cues and feedback. Although both the qualitative and quantitative results indicate that GAUdiS was considered easy to use, the metainference found a need to further improvement in this area by clarifying the existing video instructions and providing additional ones when necessary, clarifying the wording where needed, and reducing the number of clicks and amount of scrolling required. It was also found that there was a need to better consider the learning process. This can be achieved by providing more information regarding how to use GAUdiS when encountering it for the first time, informing the user about why different features are included, and allowing the user to go back and make changes in the feature for goalsetting and planning. The metainference also found a need for GAUdiS to improve the cues and feedback delivered to the user. GAUdiS was designed to send users a daily message, including reminders for goal follow-up; however, a bug prevented these messages from being delivered. The metainference highlights the importance of addressing this issue. Also, the feedback for goal follow-up was revised to provide the user with clearer information regarding changes in ST.

No evidence of an effect of the eHealth intervention (Study IV)

Participants' use of GAUdiS varied widely, with the usage range spanning from 1 to 63 active days. Median use was 6.5 active days (IQR 3-10.5). Accelerometer processing detected non-wear in one participant in the intervention group (15.27 minutes) and one participant in the control group (97.52 minutes) at baseline. No non-wear was detected at follow-up.

GAUdiS had no effect on the primary outcome, changes in ST as a percentage of wake-wear time (time* group interaction p=0.11) or on secondary outcome measures (Table 5). The LMM analysis resulted in p-values <0.05 regarding the secondary outcome measures exercise self-efficacy (time* group interaction p=0.03) and self-rated health (time group interaction p= 0.01). However, Bonferroni adjustment of the p-values for exercise self-efficacy and self-rated health showed no significant effect (p=0.33, p=0.15).

Table 5. Model-based means of primary and secondary outcome measures including 95% CI and *p*-value from the linear mixed-models analysis. [a] indicates p-values adjusted using the Bonferroni adjustment.

		INTERVENTION GROUP			CONTROL GROUP			
		n	mean	95% CI	*n*	mean	95% CI	*p* value Time* Group effect
SEDENTARY TIME % OF WWT								
	Baseline	20	64.44	57.68 to 71.21	19	64.34	57.58 to 71.09	
	Change at follow up	15	1.72	1.55 to 1.90	16	-0.89	-1.05 to -0.73	0.11
SELF-REPORTED SEDENTARY TIME LASA								
	Baseline	20	729	637 to 821	18	688	591 to 781	
	Change at follow up	15	-162	-176 to -148	16	43	29 to 53	0.07
PHYSICAL ACTIVITY IPAQ								
	Baseline	18	2500	1671 to 3328	17	2444	1565 to 3324	
	Change at follow up	13	902	620 to 1187	13	669	487 to 849	0.81
SELF-RATED HEALTH EQ VAS, EQ-5D-5L								
	Baseline	20	76.73	73.84 to 79.63	19	77.76	74.78 to 80.73	
	Change at follow up	15	8.94	8.53 to 9.34	16	-0.20	-0.44 to 0.04	0.15[a]
PERCEIVED HEALTH-RELATED QUALITY OF LIFE BBQ TOTAL								
	Baseline	20	71.51	67.72 to 75.30	19	71.68	67.79 to 75.57	
	Change at follow up	15	-0.39	-0.97 to 0.19	16	6.72	6.37 to 7.08	0.10
SUBJECTIVE WELL-BEING GP-CORE								
	Baseline	20	0.83	0.706 to 0.962	19	0.83	0.697 to 0.960	
	Change at follow up	15	-0.11	-0.13 to -0.09	16	-0.08	-0.09 to -0.07	0.83
ANXIETY HADS-A								
	Baseline	20	3.28	2.52 to 4.05	19	3.52	2.73 to 4.30	
	Change at follow up	15	-0.52	-0.63 to -0.40	16	-0.51	-0.58 to -0.44	0.99
DEPRESSION HADS-D								
	Baseline	20	1.54	1.12 to 1.95	19	1.64	1.21 to 2.06	
	Change at follow up	15	0.07	0.01 to 0.14	16	-0.51	-0.55 to -0.48	0.20
ABILITY FOR PHYSICAL ACTIVITIES								
	Baseline	20	36.31	34.91 to 37.70	19	36.36	34.93 to 37.79	
	Change at follow up	15	0.80	0.58 to 1.01	16	0.05	-0.08 to 0.18	0.62

		INTERVENTION GROUP			CONTROL GROUP			
		n	mean	95% CI	*n*	mean	95% CI	*p* value Time* Group effect
EXERCISE SELF-EFFICACY ESES-S								
	Baseline	20	28.74	25.72 to 31.76	19	29.16	26.06 to 32.26	
	Change at follow up	15	5.35	4.85 to 5.84	16	-2.11	-2.42 to -1.81	0.33[a]
SELF-EFFICACY FOR REDUCING ST								
	Baseline	20	7.99	7.23 to 8.75	19	7.88	7.10 to 8.66	
	Change at follow up	15	-0.50	-0.61 to -0.38	16	-0.62	-0.69 to -0.56	0.88
IMPORTANCE OF REDUCING ST								
	Baseline	20	8.12	7.26 to 8.97	19	7.68	6.81 to 8.56	
	Change at follow up	15	-0.43	-0.57 to -0.30	16	-0.56	-0.63 to -0.48	0.89
MOTIVATION TO REDUCE ST								
	Baseline	20	8.12	7.26 to 8.98	19	7.71	6.82 to 8.59	
	Change at follow up	15	-0.47	-0.61 to -0.34	16	-0.57	-0.65 to -0.49	0.92

Five participants in the intervention group and three in the control group withdrew from the study prior to follow-up, resulting in a 21% dropout rate. In the intervention group, the reasons given for withdrawing were: not being fond of the intervention (1); no longer needing the intervention (1); and personal reasons (3). In the control group, the reasons given for withdrawing were: not having time to participate in the study (1); and personal reasons (2). The dropout analysis showed that non-completers had greater ST than completers (p=0.04) at baseline. No other significant differences were found between completers and non-completers.

Discussion

Summary of findings

This thesis describes a health promotion project that sought to generate new knowledge on how ST can be reduced and how adherence to the reduced ST can be promoted through an eHealth intervention including self-management in OATRs. Another aim was to develop such an eHealth intervention and test its effect on adherence to reduced ST. To reach the overall aim, PD (Clemensen et al., 2017) was used as a methodological approach. Each of the four phases of the PD approach is described in a separate study: Study I describes the first phase, need assessment, in which OATRs' perceptions regarding self-management strategies for reducing ST and adhering to the reduced ST were explored by conducting focus group interviews. Study II describes the second phase, idea generation, in which workshops were conducted to explore desired features of an eHealth intervention including self-management for reducing ST and promoting adherence to the reduced ST in OATRs. In this phase, a digital prototype of such an eHealth intervention was created. In Study III – the third phase, testing and retesting the feasibility and usability of the developed eHealth intervention were examined in a mixed-methods evaluation study including five participants who used the eHealth intervention for 2 weeks. In Study IV, describing the fourth phase, evaluation, the effect of the developed eHealth intervention was evaluated in an RCT including 43 participants. Studies I and II revealed that it was important for the participating OATRs to not feel controlled and for self-management strategies to be adaptable according to their own interests and preconditions. The most important feature of an eHealth intervention to reduce ST when transitioning to retirement was considered to be support in finding joyful activities, followed by support in goalsetting and information regarding ST. Participants also emphasised that reducing ST meant making changes to their everyday life and that support is needed in order to make these changes. Affect was considered to influence reduced ST and adherence, as willingness to use self-management strategies depended on one's emotional state beforehand and on the affect that arose from using the strategy, including both joy and guilt. The wording in the eHealth intervention should be positively formulated, as this could both increase adherence and avoid feelings of guilt. While participants expressed that the order in which

the features were presented was important, they also stressed that users must choose for themselves which features to use and when to use them. The findings highlighted that it is important for an eHealth intervention aimed at reducing ST and increasing adherence to the reduced ST in OATRs to be concise, accessible and enjoyable. The developed eHealth intervention, GAUdiS, is a standalone intervention which incorporates the requested features, including one for finding joyful activities and one for goalsetting and planning based on MCII. Study III, which examined the feasibility and usability of GAUdiS, found that the intervention was modestly feasible and usable. Participants saw potential in GAUdiS's ability to reduce ST during the transition to retirement, and described the inclusion of joyful activities as an appropriate and promising approach. However, participants did not reduce their ST between phases. Feasibility was influenced by the frustration participants experienced when the eHealth intervention did not function as expected. Although GAUdiS was generally perceived as easy to use, the lack of prior external instructions placed high demands on clarity and intuitiveness. The metainference indicated that GAUdiS was modestly feasible and usable, but that further development was needed in order to strengthen both feasibility and usability. Three key areas for improvement were identified: ensuring clarity and ease of use, supporting the learning process, and providing more effective cues and feedback. Study IV, presenting the evaluation phase of this PD project, found that GAUdiS had not reduced ST at 3 month follow-up. No changes were found in the secondary outcome measures.

Understanding reduced sedentary time: How the findings and the features of GAUdiS align with – and challenge – the IBC model

Central findings discussed in this thesis involve the importance of self-management strategies for reducing ST in OATRs and of an eHealth intervention delivering these strategies, providing a feeling of being in control and not being told what to do. Other central findings are the importance of providing self-management strategies targeting both affective and cognitive determinants for reduced ST, including finding joyful activities. When interpreted in relation to the IBC model (Hagger & Chatzisarantis, 2014), these findings indicate that participants wish to experience autonomy and to be autonomously motivated in relation to the self-management strategies and the eHealth intervention.

The need for autonomy brings attention to a potential tension between central concepts in this thesis and a reduction of ST. The overarching aim of the thesis is health promotion, which seeks to empower individuals to

take control over their health (World Health Organization, 2021, p. 4). The definition of adherence emphasises agreement between the caregiver and the individual regarding the recommendations provided (World Health Organization, 2003, p. 3), with a focus on partnership and shared power (Snowden et al., 2014). The concepts of autonomy, health promotion and adherence share some foundational similarity: they all rely on individuals experiencing genuine self-determination, active involvement, and power and control over their health and health behaviour. Each concept assumes that meaningful and sustainable behaviour change can only occur in alignment with an individual's own values and preferences, and must provide a sense of agency. However, sedentary behaviour is often affectively driven in older adults (Compernolle et al., 2020; Maher & Dunton, 2019) and is frequently performed because it is expected to promote wellbeing (Niklasson et al., 2023), health (Eklund et al., 2021) and often occurs within high-value, purposeful activities (Palmer et al., 2019). This raises an important question: can autonomy truly be supported when OATRs are encouraged to change a behaviour that may contribute to their wellbeing and perceived health?

Several measures have been undertaken to meet the assumption of health promotion and adherence, and to support autonomy. The first is the inclusion in GAUdiS of features for finding joyful activities. If a person reduces their unhealthy ST, meaning time spent in unwanted ST, loneliness or boredom, finding joyful activities could theoretically increase their health. However, if ST is reduced through a reduction in ST that the individual considers to be healthy for them, it is crucial that they find joyful activities in order not to jeopardise their autonomy and health.

Furthermore, an attempt was made to achieve autonomy by giving the user the possibility to use the eHealth intervention ad libitum, in whatever way that best suited them. This was important to the participating OATRs, who called for voluntariness regarding both frequency of use and how the intervention´s features were used. Careful consideration was also taken regarding the wording in the eHealth intervention to support autonomy. Throughout the development of the eHealth intervention, it was found that the wording should not give the user feelings of guilt or that they are being lectured regarding spending time in sedentary behaviour. Thus, encouraging and positively framed language was used.

Another finding from the studies included in this thesis is the importance of including information regarding ST, and that this information should be given early so the user can understand why reducing ST is important and why they should continue to use the eHealth intervention and self-management strategies. The information includes the advantages of reducing ST, the fact that ST can be reduced with the help of small everyday changes,

information that reducing ST is not the same as starting to exercise, and the fact that undertaking enjoyable activities can be a way to reduce ST. Interpreted from the IBC model, the feature for information regarding ST targets affective and cognitive attitudes as well as implicit attitudes and motivation, both regarding sedentary behaviour and regarding joyful activities as alternative behaviours for reducing ST. The feature for information can also influence perceived behavioural control within the IBC model and the underlying concept of self-efficacy (Ajzen, 2002), as the user gets the information that ST can be reduced by making small and/or enjoyable changes.

Another important feature of an eHealth intervention to reduce ST was thought to be support for goalsetting. In GAUdiS, this feature is integrated with the feature for planning using the method MCII (Oettingen, 2012). MCII integrates well with the IBC model as MCII includes implementation intentions which is closely linked to and sometimes even overlaps with the concept of action planning, which is included in the intention–behaviour connections outlined in the IBC model. Also, when setting a goal for the first time the user is prompted to measure ST by self-assessing how sedentary they were during given time periods on the previous day. This self-assessment can lead to finding and becoming aware of cues for sedentary behaviour (Gardner et al., 2020); this can be helpful in identifying obstacles to goal achievement within MCII, associated with the concept of action planning within the IBC model as described earlier. The self-assessment also strives to increased perceived behavioural control through increased self-efficacy (Michie et al., 2013) for reduced ST. Further, MCII can influence both implicit attitudes towards the target behaviour (Oettingen et al., 2001) as well as implicit attitudes towards one's obstacles (Wittleder, 2020). As the present thesis targets affect when using MCII, it is reasonable to assume that implicit affective attitudes and implicit motivation is influenced.

As demonstrated above, the self-management strategies for reducing ST that emerged from the findings align with and target concepts in the IBC model. Moreover, every concept within the model is addressed, either directly or indirectly, through the design and features of the developed eHealth intervention GAUdiS. This indicates that the IBC model is suitable to use when designing eHealth interventions to reduce ST in OATRs. However, the absence of determinants targeting alternative behaviours in the model is a problem, and the inclusion of alternative behaviours would further enhance the model's applicability in interventions to reduce ST.

The IBC model was developed to target the health behaviour of physical activity. ST and physical activity differ in several aspects, a major one being the matter of behaviour change. When addressing ST, it is a matter of behavioural reduction instead of behavioural increase, which is most often the case regarding physical activity. In this respect, reducing ST by finding

joyful activities is in this respect a matter of habit substitution (Gardner et al., 2023). When using habit substitution to reduce a non-desired behaviour, it is advocated that one use appropriate interventions to increase the alternative behaviour in order to achieve behaviour change. Also, throughout the findings in this thesis it is clear that self-management strategies for reducing ST should not only target a reduction of time spent in sedentary behaviour (the target behaviour) but also an increase in joyful activities (alternative behaviours).

As I see it, this could be done in two different ways. One way is to add a new concept to all the existing ones within the IBC model, which would target alternative behaviours. However, this would mean the inclusion of a great number of concepts. Another model, regarding affective determinants for health behaviour (Williams & Evans, 2014) includes the concept 'motivation for competing behaviours'. However, this approach would not be suitable for reducing ST in OATRs as it would indicate that alternative behaviours influence ST but would fail to illustrate determinants of the alternative behaviour. As described earlier, determinants (both cognitive and affective) for alternative behaviours are important for reducing ST in OATRs. Instead, it could be clarified that the existing concepts within the IBC model address both the target behaviour as well as alternative behaviours. Such an addition would incorporate the determinants of alternative behaviours and, in doing so, render the model more suitable for explaining reduced ST.

Lastly, as the results of this thesis highlight the importance of both cognitive and affective determinants when addressing reduced ST in OATRs a clarification that attitudes include both affective and cognitive elements would further strengthen the use of the model when designing such an eHealth intervention.

Active components of GAUdiS: Contributions to reduced sedentary time and potential pitfalls

As the name of the developed eHealth intervention is GAUdiS – 'Joyful activities facilitate reduced ST' a central question is whether or not ST can be reduced by finding joyful activities. In the results of Study IV, the evaluation phase, the analysis showed that GAUdiS did not lead to reduced ST. This may indicate that the intervention is unable to support reduced ST in the transition from working life to retirement. However, it does not necessarily mean that ST cannot be reduced in this transition through finding joyful activities. A cross-sectional study of 595 participants aged 66 years or older found that social participation was strongly associated with less time in ST and more time in light physical activity (Chen et al., 2021). In

comparison with other research, it is important to consider that in this thesis the concept of joyful activities is a broad concept that is not limited to participating in social activities with others in society. Instead, a joyful activity is regarded as any activity performed throughout the day that gives one a feeling of joy. This can come from undertaking an activity one considers fun, changing how one perform an everyday activity to be more joyful, doing something with or for someone else, or making oneself feel meaningful. In Study IV evaluating the effect of GAUdiS, the participants' use of GAUdiS is reported, but not whether or not this usage resulted in increased engagement in joyful activities. As it is unknown whether participants in the intervention group increased their engagement in enjoyable activities, Study IV cannot be considered a conclusive test of their potential effect on ST. Further, in Study II, participants suggested that the eHealth intervention could send notifications about potential joy-giving activities available nearby. This was considered difficult to implement due to the time required; however, with current technology, such functionality might be feasible through an AI-based solution. Further research on the feature for identifying joy-giving activities is therefore proposed to enable its use in reducing ST.

The other main component of GAUdiS is the feature for goalsetting and planning based on MCII (Oettingen, 2012; Oettingen et al., 2001). No earlier studies have tested the effect of MCII on reducing ST. MCII has been successfully used to change other health behaviours (Cross & Sheffield, 2019; Ruissen et al., 2018), including using eHealth interventions (Wittleder et al., 2019). However, there is a risk that the inclusion of MCII could undermine the effectiveness of GAUdiS in reducing the participants' ST. In MCII, the first two steps are to identify and define a meaningful goal, followed by imagining the desired outcome of achieving that goal – in this case the affective outcome, meaning how one would feel and what affect would arise from reducing ST. This assumes that the user is autonomously motivated to reduce ST, considers a reduction in ST to be a personally meaningful goal, and anticipates positive affective outcomes from the goal achievement. As older adults are often unaware of how much time they spend being sedentary and of the health risks associated with excessive ST (Compernolle et al., 2020; McGowan et al., 2021), and ca experience sedentary behaviour as meaningful and healthy (Eklund et al., 2021; Palmer et al., 2019), it is possible that they do not perceive it as an important goal for them. If this is the case, they may find it difficult to recognise and articulate any positive affect they would experience from reducing their ST, which in turn would limit the implicit motivational processes that the MCII method is intended to activate. To counteract this risk, the user is provided with information regarding ST when logging in to the eHealth intervention and in the daily message sent to their phone. This information is intended to make the user aware of

the benefits of reducing ST, to clarify that reducing ST does not necessarily require starting to exercise but can instead involve engaging in joyful activities, and to emphasise that it can be done in their own way. This is intended to support users' autonomous motivation to reduce ST.

There is also a risk of GAUdiS undermining a reduction in ST, linked to the feature for ST measurement that users carry out prior to setting a goal and again at goal follow-up. In this feature, the participants are asked to self-assess how sedentary they were during the preceding day. The feature for measuring ST was considered important by the participants in study II as they expressed that the user must get an idea of their daily ST to know if it is relevant for them to use the eHealth intervention and reduce ST and for the user to be able to set relevant goals. Although objective measurement of ST would provide greater precision (Prince et al., 2020), did participants in Study II prefer self-assessment so that the eHealth intervention would not incur additional costs and they would not have to carry a device at all times. Although asking about the previous day is recommended for self-assessed ST, there is a risk of both overestimation and underestimation when making self-assessments regarding ST (Prince et al., 2020). If this self-assessment is inaccurate at goal formulation or follow-up, there is a risk that the user will set an ST goal that is too difficult to achieve, or wrongfully believe that they did not achieve the goal. When using MCII, goal commitment is strengthened when self-efficacy is high and obstacles are perceived as possible to overcome, whereas low self-efficacy and perceiving obstacles as insurmountable weakens goal commitment (Oettingen, 2012; Oettingen et al., 2001). Thus, the combination of incorrectly assessed ST followed by MCII could lead to weakened goal commitment.

GAUdiS also provides the user with tips for reducing ST. These tips, which come from OATRs were collected from data from Studies I and II. The inclusion of some of these tips – such as planning fun activities to do at home, listening to music or an audio book while moving around, making a list of past interests to try again, standing up when talking on the phone, or taking out the garbage as soon as the bag is full instead of waiting until one is going out anyway – can be discussed, as there is no evidence that these tips will reduce ST. However, participants described them as ways to reduce ST in transition to retirement. Before being incorporated into the intervention, they were carefully reviewed to ensure that they were considered reasonable for reducing ST and were not judged to pose a risk to the participants. Furthermore, as part of the methodological approach of this thesis, it is essential to empower the OATRs and to value their knowledge as equal to that of the researchers. It is therefore important to consider their perspectives on how ST can be reduced. Their insights are particularly crucial, as they possess a unique understanding and experience of their own everyday

lives. Additionally, some of the tips they provided for example being active when meeting friends and family or dancing to one's favourite music, are also included in the Public Health Agency of Sweden's suggestions for reducing ST (Hellenius, 2022). The tips can also be understood as suggestions for habit disruption and ways of overcoming barriers, which are well-established strategies for influencing habit (Gardner et al., 2020).

As described earlier, GAUdiS delivers a daily message to the user's telephone. These messages include the information previously outlined, a reminder of the time for goal follow-up or one of the previously described tips for reducing ST. The results from Study III the test and retest phase revealed that it was considered suitable to receive one message per day at a time chosen by the user. With the current technology, there are extensive possibilities to deliver just-in-time (Nahum-Shani et al., 2018) prompts and messages when individuals are exposed to cues, are sedentary, or have been sedentary for an extended period. Today's phones and smart watches can monitor ST and give notifications when one has been sedentary for a prolonged time. However, studies of such just-in-time interventions have shown mixed results regarding their ability to reduce ST (Longhini et al., 2024), and an intervention targeting OATRs was not found to be effective (Suorsa et al., 2022). Further, the results of Study I and II indicate that although participants expressed that self-monitoring using technical devices and notifications when spending extended time in sedentary behaviour offered potential in reducing ST, they perceived these approaches as overly controlling and thus as counteracting their autonomy.

The participating OATRs expressed that the features should be presented to the user in a certain order but emphasised the importance of voluntariness regarding when to use the eHealth intervention and which features to engage with. This is in line with earlier research, which found that a tunnelled design in which the user had less control was experienced as less efficient by participants; however, the same study found that this tunnelled design resulted in more pages read, more time on the website and more gained knowledge compared to a freedom-of-choice version (Crutzen et al., 2012). In GAUdiS, the feature for goalsetting and planning is tunnelled to provide the user with the requested support in the order articulated by the participants in the workshops in Study II. However, to support autonomy, the user can choose when and even whether to use the features. There is a risk that the flexibility to use GAUdiS freely may have led to the participants not engaging with all of its features. As a result, participants may not fully have assimilated the information regarding sedentary behaviour when they began using GAUdiS – a feature that participants emphasised as important during the development (Study II), and which previous research has identified as a contributing component in reducing ST (Petrusevski et al., 2020).

Participants in the first two phases (Studies I & II) expressed that a digital solution can be suitable for reducing ST but that it is not appropriate for everyone, and that some OATRs might prefer an in-person intervention. The decision to use an standalone eHealth intervention was made because the societal challenges we currently face, with an ageing population (World Health Organization, 2015), require health promotion interventions that can reach large numbers of older adults and be accessed outside the healthcare system. However, eHealth interventions for reducing ST in older adults and adults often include in-person meetings or personal contact via telephone (Yerrakalva et al., 2019; Zhang et al., 2022) and research on other health behaviours have found standalone eHealth interventions to be less effective than those including professional contact (Kupila et al., 2023). Therefore, future research could explore the addition of an intervention delivered outside the eHealth intervention, thereby creating a hybrid solution.

Enhancing the possible health effects of reducing sedentary time with GAUdiS

In this thesis the focus has been on reducing total daily ST. Earlier research has focused on other aspects of ST such as sedentary bouts (Wilson et al., 2021) or screen time (Hudgins et al., 2023). A central question is how health promotion interventions should modify sedentary behaviour in order to achieve the greatest health impact. As described earlier, previous research has found that sedentary behaviour can be experienced as both healthy and unhealthy (Eklund et al., 2021), and that there is both high-value and low-value ST (Palmer et al., 2019). Greater health benefits may be achieved by shifting the focus from total ST to the time an individual perceives to be unwanted and unhealthy sedentary behaviour. Such a focus could enhance individuals' autonomous motivation for reducing their ST as they would have the power to decide for themselves which ST to reduce. This focus would also be more consistent with the underlying values of health promotion, adherence and health described earlier – namely, supporting individuals in experiencing genuine self-determination, active involvement and a sense of power and control over their health and health behaviours.

Furthermore, replacing unhealthy ST with joyful activities could have other health benefits. A longitudinal study including 93,263 older adults from 16 nations found that engaging in a hobby is associated with fewer depressive symptoms and higher self-reported health, happiness and life satisfaction (Mak et al., 2023). In Study IV, the analysis of the secondary outcomes found a nonsignificant increase in self-rated health. This might indicate that the participants have found new activities in their everyday life and

thereby experience increased self-rated health. More research is needed to determine whether GAUdiS has an effect on self-rated health; in such research it would be of interest to examine whether participants do engage in more joyful activities due to their use of GAUdiS. Lastly, it would be of interest to examine the potential health effects of substituting unwanted sedentary behaviour with joyful activities.

Methodological considerations

This thesis uses PD (Clemensen et al., 2017) as its methodological approach. The approach was chosen as part of the aim of the thesis was to develop an eHealth intervention. PD offers a way to involve end users as active participants in the design process when developing new eHealth interventions (Clemensen et al., 2017). When designing such interventions it is crucial to include end users in the design process as they possess unique lived experience regarding their everyday life, their barriers and their opportunities. In the present thesis targeting OATRs, end-user participation is of the utmost importance as they represent a group that is at risk of digital exclusion (Pérez-Escolar & Canet, 2022). It is therefore essential to develop an eHealth intervention that is both accessible and easy to use.

Moreover, PD was an appropriate methodological choice, because it shares core values with the underlying concepts of this thesis. Consistent with the core values underpinning health promotion (World Health Organization, 2021, p. 4), the definition of health (World Health Organization, 1948, p. 1), person-centredness (Ekman, 2014), and adherence (Snowden et al., 2014; World Health Organization, 2003, p. 3), it emphasis shared power, participation and empowerment.

PD has its roots in action research (Reason & Bradbury, 2008) and PR (Vaughn & Jacquez, 2020), in which end-user involvement across all research phases is considered the gold standard, and where the quality of the research is often evaluated based on the extent to which participants have been involved in every stage from planning to analysis and manuscript preparation (Khodyakov et al., 2013) as well as by the level of this involvement (Fischer et al., 2020). In the present thesis OATRs were involved as participants throughout the development of an eHealth intervention using PD, although their level of active participation varied between phases. Focus groups interviews and workshops were carried out in the first and second phases, need assessment and idea generation, described in Studies I and II, which may be interpreted as a high level of participation (Fischer et al., 2020). In the third phase, testing and retesting, described in Study III, the participants participated by testing the developed eHealth intervention and

they had the opportunity to give direct input to guide its further development, which may be considered an intermediate level of participation (Fischer et al., 2020). While conducting workshops in Study III would have increased the level of participant involvement for that phase, this option was not pursued; as the development was considered to be at a later stage in the process, workshops had already been conducted in Study II, and the aims of Study III could be achieved without additional workshops. Additional workshops would also have added substantial time demands. While study IV does not involve participatory elements when examined separately, it plays a crucial role within the PD project by addressing the final phase, evaluating the intervention's effectiveness, typically achieved through an RCT. Overall, the degree of participation within this PD project can be classified as intermediate to high as the participants had an active role in the intervention's development and had the opportunity to directly influence this development (Fischer et al., 2020).

The OATRs were, not actively involved in developing the research idea or the aims of the project, however. Given the project's value base, which emphasises that individuals should have the power to decide over their own health, it might have been reasonable to invite the OATRs into a dialogue. Such a dialogue could have focused on which health behaviours they themselves would like to change in order to best improve their health. Compared to other health behaviours such as physical activity or smoking, sedentary behaviour is a relatively novel concept, and older adults are often still unaware of the health benefits of reducing ST (Compernolle et al., 2020; McGowan et al., 2021). Consequently, if OATRs had been involved in the development of the research question there would have been a risk that sedentary behaviour would be missed as the health disadvantages of other unhealthy behaviours are more established among the public. The consequences of this could have been an intervention that does not primarily target ST specifically, and such interventions has been found to be less effective in reducing ST (Prince et al., 2014). Given the evidence on the health disadvantages of high levels of ST (Ajufo et al., 2025; Biswas et al., 2015; Ekelund et al., 2019; Rossen et al., 2020), it was valuable to direct this health promotion project specifically towards a reduction of ST.

In contrast to PD, eHealth interventions can be developed with guidance from a more explicitly theory-driven approach like Intervention mapping (Fernandez et al., 2019). There are no clear boundaries between theory-driven and participatory approaches to eHealth development, as the development of eHealth interventions often incorporates elements from both. Instead, the chosen methodology illustrates where the main emphasis of the project lies between the two approaches. PD rests on the fundamental value that all participants' knowledge is regarded as equally important. In this

thesis, two of the participants in the workshops conducted in Study II, were researchers within physiotherapy with a behavioural medicine approach, and contributed their knowledge of behaviour change processes and theories explaining health behaviour. Consequently, academic knowledge about relevant theories was included and treated as equally important as the participating OATRs' perspectives.

Rather than deciding on a theory prior to the workshops, a theory to explain the components of the developed eHealth intervention and understand reduced ST in OATRs was discussed at the end of the second phase, idea generation (study II). This was done after careful consideration, as it was deemed that introducing a theory at too early a stage could undermine the participatory foundation of power-sharing by placing too much decision-making authority in the hands of the researchers. Moreover, as sedentary behaviour, is a relatively new field of research, and little is known about how ST can be reduced in OATRs, deciding on a theory prematurely could have steered the discussions and outcomes in ways that might have prevented new knowledge from emerging. For example, the prioritization of the features and the importance of including features for finding joyful activities, for goal setting and planning, and for information, was not known beforehand. A predefined theory could therefore have steered the discussions away from one or more of these features. It can be debated how such a steering might have influenced the effectiveness of the intervention, but it would have risked departing from the participatory foundation and might have resulted in a theoretically structured intervention that would not fully accommodate the needs and preferences of OATRs.

However, the use of PD does not guarantee the development of an effective and accepted eHealth intervention. There is limited research regarding whether PD leads to more effective eHealth interventions but there is some research regarding the effect on implementation outcomes. A systematic review of the involvement of older adults in technology design found that the evidence regarding whether involving older adults in the design process benefits acceptance and adoption is inconclusive, with studies revealing both positive and negative effects (Fischer et al., 2020). Within this PD project, deliberate measures were taken to support and enhance the positive effects that can arise from involving participants in the design process. One of these measures was to emphasise the participants' different roles during the workshops (Denecke et al., 2025), and to critically examine the data during analysis to ensure that only data relevant to each participant's role were included. For example, statements about how the web developer or the researchers themselves wished to reduce their own ST would not have been included in the analysis, as such data did not fall within their participant roles. This approach could have been further applied in the evaluation of the

different features by specifying the roles of the individuals who made each assessment. Another measure that was taken to enhance a possible positive effect of involving OATRs was to ensure that the participating OATRs were referring to what they wanted for themselves rather than what they thought others would want. This approach has been advocated in prior research (Yardley et al., 2015). However, in the focus group interviews conducted in Study I, participants were asked how they would advise someone else to reduce their ST. This approach was used to initiate the discussion: as sedentary behaviour can be perceived as something negative among older adults (McGowan et al., 2021), and as participants in Study I might not experience that they needed to reduce their ST, the researchers hypothesised that they would find it easier to start talking about ST when the focus was on others rather than themselves. However, the discussion then evolved into reflections on how they themselves would apply, or already had applied these methods to reduce their ST. As such, this is not considered to have adversely influenced the results.

Finally, the workshops in Study II were planned to facilitate active participation. For instance, established PD tools were used (Vandekerckhove et al., 2020), such as cards, 2D mapping and user journey were used, which is considered to have encouraged engagement among all participants and to have resulted in a high level of participation. However, adding digital tools rather than a paper prototype at the third workshop could have supported even greater participation by offering opportunities for more hands-on design work, as has been done in earlier research (Revenäs et al., 2015).

There is a risk of selection bias (Polit & Beck, 2021) in this thesis as those with an interest in the use of digital devices or eHealth might be more likely to enrol in the studies included. Further, participants were recruited from retirement organisations and theme days for all the studies (I-IV). This may have influenced the findings in Studies I and II and the developed eHealth intervention, if those who participated in the studies had a greater interest in joining a retirement organisation and therefore tended to prefer this type of activity. There is also a risk that this recruitment strategy might have influenced the results of Studies III and IV. As participants in Study I and II described having reduced their ST by finding joyful activities, and participants in study III and IV were recruited using the same strategy, there is a risk that participants in Studies III and IV also already used joyful activities as a self-management strategy for reducing their ST, which may have influenced the extent to which the study was able to detect an effect of the intervention. To counteract the risk that participants were exclusively individuals with an interest in retirement organisations, participants were also recruited through information on municipalities' digital intranets for employees (study I, III-IV), by advertising on social media (Studies III-IV), and via

companies, with a request to spread the information to their employees (Study IV).

In Studies III and IV the additional inclusion criterion of spending an average of 6 hours of daily ST was added to ensure that participants were those who needed to reducing their ST. The cut-off of 6-8 hours ST has been used in earlier research (Gardner et al., 2014; Patterson et al., 2018), but could be discussed as other research has found that the health risks associated with ST increases at 9.5 hours per day (Ekelund et al., 2019). Changing the inclusion criterion to only include those who have ST of 9.5 hours would have included those who would benefit the most from reducing their ST; however, recruiting participants for health intervention studies is a challenge (Axén et al., 2021). This was also the case in the recruitment for the studies in this thesis, with recruitment for Studies III and IV becoming even more challenging due to the COVID pandemic. Although the recruitment took place after the pandemic, retirement organisations describe a decline in participation and a reduced recruitment of those who are newly retired, which made it harder to reach potential participants. Recruitment might also have been challenged by older adults often being unaware of their ST (Compernolle et al., 2020). Potential participants might therefore have underestimated their own ST and consequently declined to participate. Thus, as raising the ST threshold for study inclusion would consequently have made recruitment even more challenging, the inclusion criterion of 6 hours daily ST was appropriate.

The participating OATRs were predominantly women. In, Studies III and IV the majority had a current or past mostly sedentary occupation, and a slightly larger proportion were highly educated. This is important to consider when reflecting on the transferability of the findings in Studies I-III and concerning the generalisability of Study IV. The difficulty recruiting older men for health promotion interventions is well known (Howell et al., 2023). The recruitment from retirement organisations may have contributed to the underrepresentation of male participants, as more women than men are members of retirement organisations. A more targeted recruitment strategy might have increased the number of male participants. A potential future strategy would be to engage male OATRs in discussions of how recruitment approaches could be shaped to enhance male participation (Howell et al., 2023).

In Study I-III which were based on a qualitative paradigm, qualitative content analysis (Graneheim & Lundman, 2004) and TA (Braun & Clarke, 2021) were used – two well-established analytical methods when the aim is to analyse participants' perceptions (Braun & Clarke, 2006; Braun & Clarke, 2021; Lindgren et al., 2020). In these studies, measures were taken to enhance the trustworthiness (Lincoln & Guba, 2007) of the findings. A

selection of these measures included increasing credibility by pilot testing of the interview guide in Study I and using the feature analysis and member checking as verifications of the paper prototype in Study II. The study procedures of Studies I-III were well planned and documented and in order to increase dependability. Also, quotations were included in the results of Studies I-III to enhance credibility and confirmability. Finally, subcategories, categories (Study I), and themes (Study I-II) were discussed among all authors of the studies, and the use of special software enabled the possibility back and forth between themes, categories, meaning units, codes, and the original audio recordings, thereby strengthening credibility and confirmability.

In Study III, which described the phase; testing and retesting, the feasibility and usability of GAUdiS were examined. Careful consideration was made to understand and describe what feasibility is and how it can be examined. Concepts linked to feasibility, such as acceptability and usability, tend to overlap and are in many ways connected, and it can be difficult to draw a line regarding the difference between the concepts. In this thesis, feasibility is understood as the intertwined outcome of acceptability and participants' response, meaning the thoughts and affect the user experiences when engaging with the eHealth intervention (Perski & Short, 2021), and whether use leads to behaviour change (in this thesis, changes in ST). However, evaluating acceptability of an eHealth intervention for behaviour change is a matter of not only evaluating the intervention itself, but also the acceptability of the components for behaviour change included in it (Yardley et al., 2015) which was an important aspect in Study III.

As SCED (Krasny-Pacini & Evans, 2018) was chosen to evaluate changes in ST in Study III, five participants were included. The limited number of participants may seem to be a threat to the credibility of the qualitative findings. However, the number of participants was deemed adequate, given that the research question was not considered to be particularly profound or complex. Moreover, a similar number of participants has been used to assess feasibility using SCED in previous research (Schrijnemaekers et al., 2020). Furthermore, the use of SCED is widely applied in the evaluation of health behaviour interventions and has been endorsed in earlier research (Krasny-Pacini & Evans, 2018). The SCED data were analysed using both visual and statistical methods, consistent with recommended procedures for SCED analysis (Krasny-Pacini & Evans, 2018; Kratochwill et al., 2010). For statistical analysis TAU-U was used which is appropriate in data with highly variability and can correct for trend (Parker et al., 2011). This together strengthens the interpretation of participant response measured as changes in ST in Study III.

In Studies III and IV, accelerometers were used to measure ST. Accelerometer measurement is often highlighted as the benchmark approach for assessing ST, rather than using self-assessment. Both the accelerometer and the algorithm used for collecting and processing data have been validated for assessing ST (Hedayatrad et al., 2021; Skotte et al., 2014; Stemland et al., 2015) which is a strength of the studies. Still, the use of accelerometer data necessitates decisions about data processing, and the act of measurement may itself influence participant behaviour. One data decision concerns how to handle non-wear and how much of it can be tolerated over the course of a day for that day to be considered valid. Studies III and IV applied the commonly used criteria of at least 20 hours of wear time, one walking period, and any duration of sleep during a 24-hour period for a day to be considered valid for inclusion (Blodgett et al., 2024). However, this allows for up to four hours of non-wear time, creating a time window during which participants' ST is unknown and thus, threatening the validity of the findings. In Study III mean wear time was 23 hours 58 minutes, and non-wear was found on 3 days (29.72-91.12 minutes). In Study IV non-wear was found on one included day for one participant in the intervention group (15.27 minutes) and one participant in the control group (97.52 minutes). Thereby, wear time was considered high, increasing the validity of the results. The high wear time was probably due to the accelerometer being placed on the participant's thigh with a plaster, thereby eliminating the risk of their forgetting to reattach the accelerometer, which is a common issue when devices are removed overnight. Another concern to address when analysing accelerometer data is how to handle differences in wake-wear time caused by differences in sleep. There is no consensus on how differences in sleep should be handled when analysing accelerometer data. In Studies III and IV it was addressed by presenting ST as ST in percentage of wake-wear time. With this methodological approach, both non-wear time and differences in sleep were handled. There is also a behaviour-related issue regarding measurement reactivity when assessing ST using accelerometers, namely that participants tend to change their behaviour and reduce their ST due to wearing the accelerometer (Ullrich et al., 2021). In Studies III and IV, measurement reactivity was addressed by encouraging participants to maintain their usual daily routines. Also, ultra-sensitive adhesive patches were used, leading to participants reporting that they did not notice wearing the accelerometer; this may have reduced the impact of measurement reactivity as participants might have forgotten that they were wearing the accelerometer.

In Study IV, LMM was used to examine differences in primary and secondary outcomes between the intervention and control group at the 3-month follow-up. LMM is a commonly used analytical approach for evaluating the

effects of interventions aimed at reducing ST (Kroesen et al., 2024; Kroesen et al., 2025; Suorsa et al., 2022). However, another analytical method that has been used in ST research is compositional data analysis (Dumuid et al., 2020). A key advantage of compositional data analysis is that it can identify which behaviours replace ST when a reduction occurs. Although this was beyond the scope of Study IV, future research using compositional data analysis could provide additional insights into how ST changes from a 24-hour perspective. Such an analytical approach would also account for variations in sleep across measurement occasions.

Ethical discussion

There are important ethical considerations to take into account in a health promotion project aimed at reducing ST through an eHealth intervention among OATRs. First, some considerations related to delivering the intervention as an eHealth intervention can be mentioned. It is important to reflect on which groups might be excluded by choosing an eHealth format for the intervention. Despite the increasing use of digital devices among OATRs, there are still individuals who do not wish to use a digital solution, and it is important to consider digital exclusion among older adults (Pérez-Escolar & Canet, 2022). In this project, PD (Clemensen et al., 2017) was used as a methodological framework with the goal of creating an eHealth intervention in line with the end users needs. Also, the inclusion of a feasibility study (Study III) aimed at further refining the eHealth intervention to be feasible and usable for OATRs. From an ethical perspective, it is important to also consider whether the intervention could be delivered in a non-digital version as well. This could be developed in future studies, although many of the functions available in a digital solution, such as tailoring and messages, would not be possible to provide in a non-digital version.

There are also considerations to take into account related to health promotion. In health promotion interventions, there is a risk of inadvertently creating feelings of guilt among individuals about their current behaviour through blaming and stigmatisation (Tengland, 2016). These feelings would counteract the health promotion intention of empowerment and may also undermine the user's sense of autonomy. In Studies I and II, participants discussed the importance of the wording of the information in the eHealth intervention not evoking feelings of guilt in users. Accordingly, all wording in the eHealth intervention was thoroughly reviewed to seek to ensure that it did not inadvertently evoke feelings of guilt in users. It is also important to remember that sedentary behaviour can be unwanted, for example in connection to loneliness or having nothing to do. From an ethical perspective,

it is therefore important to offer alternatives to sedentary behaviour so that users of the eHealth intervention feel that they have the opportunity to follow the recommendations provided and do not experience feelings of sadness or discouragement when using the intervention.

In all research it is essential to reflect on which individuals are excluded from participation, and thus whose voices are not heard or represented in the findings. The studies in this project excluded those who were unable to speak and understand Swedish or to read and comprehend the study instructions, as well as those who had a serious disease, severe loss of vision, or severe loss of communicative ability. From an ethical perspective, future research should specifically target, for example, those who do not speak Swedish fluently, individuals with various severe illnesses, and people with different disabilities, in order to develop interventions for them through PD.

Conclusion

This thesis has highlighted the importance of supporting autonomy when designing eHealth interventions for reducing ST and promoting adherence to the reduced ST in OATRs. Participants wanted to feel that they were acting on their own will and according to their own interests, and did not want to feel controlled. The strive for autonomy was found to be related both to how participants wanted self-management strategies to be adaptable to their own interests and circumstances and to not make them feel controlled, and to their desire to decide for themselves how to use the eHealth intervention rather than being directed in how it should be used.

Further, this thesis indicates that self-management strategies for supporting OATRs in reducing their ST and increasing adherence to the reduced ST should target both affective and cognitive determinants. Finding and engaging in joyful activities to fill one's newfound free time was considered an important self-management strategy for reducing ST and for adhering to the reduced ST. At the same time, changes in everyday life were required in order to achieve a reduction in ST, and participants expressed a need for support in making these changes. The three most important features for an eHealth intervention targeting reduced ST in OATRs were considered to be finding joyful activities, goalsetting and information regarding ST.

Also, it is suggested in this thesis that the IBC model can be used to explain how features included in the developed eHealth intervention GAUdiS strive to target determinants in order to reduce ST. Further development of the IBC model is suggested to better account for sedentary behaviour as a habit-driven behaviour and to clarify the inclusion of both affective and cognitive determinants. In line with the findings of this thesis, such development may also involve explicitly addressing alternative behaviours. This may represent a pathway to support adherence, as the findings indicated that autonomous motivation towards these alternatives was considered important for reducing ST.

Although the eHealth intervention was developed with OATRs using PD, in which the OATRs were considered to be active and to have high influence, it was considered to be modestly feasible and usable. Although feasibility and usability were influenced by bugs that occurred, additional needs for further development were also identified. This was partly due to the high

demands placed on clarity and simplicity, given that the eHealth intervention is a standalone solution.

The developed eHealth intervention did not show any effect on ST. Based on the results of this thesis, it cannot be concluded whether joyful activities or MCII are effective self-management strategies to reduce ST in OATRs, as the evaluation phase (Study IV) included a small number of participants and the extent to which participants interacted with the different features of the intervention is unknown. Additional components beyond those offered by the current version of GAUdiS may be required to reduce ST in OATRs.

Practical implications and future directions

As we arrive at the end of this adventure, it is time to reflect on how the knowledge generated may inform future practice and which insights may guide further scientific inquiry. Although the results of Study IV showed that the developed eHealth intervention did not reduce ST, new knowledge has been generated regarding self-management strategies and eHealth interventions for reducing ST and increasing adherence to the reduced ST in OATRs.

For healthcare professionals who meet OATRs in their practice and who therefore address the issue of sedentary behaviour, the findings of this thesis may be applied by recognising the importance, as expressed by participants, of supporting autonomy when discussing ST during consultations. Healthcare professionals may also consider affective determinants of both sedentary behaviour and alternative behaviours, to support reductions in ST. In addition, participants expressed a need for support in changing everyday life, particularly with regard to goal setting and receiving positively framed information about reducing ST. When incorporating eHealth, it may be important to consider that support beyond what can be provided by a standalone eHealth intervention may be required to effectively reduce ST in OATRs.

In all great adventures there is always a sequel. While the aims of the thesis have been fulfilled, new questions have emerged during our journey, which highlights the need for further research. Several directions for future research have been outlined in the discussion, and a few summarising points will be highlighted here.

First, as the eHealth intervention did not demonstrate an effect, further development may be warranted. Such development could include the adoption of a hybrid intervention approach, as participants expressed that a standalone eHealth intervention may not be suitable for everyone, and as previous research on eHealth interventions has often employed hybrid approaches. Further research using PD is therefore called for to examine how ST can be reduced through hybrid interventions using GAUdiS.

Moreover, the rapid development within the digital field, particularly advancements in artificial intelligence, is creating opportunities to further enhance the eHealth intervention and more effectively tailor it to users. In

Study II, participants expressed a desire not only for suggestions for nearby activities, which are included in the current version, but also for notifications about joyful activities that might be of interest on a given day. This could not be included in the current version of the intervention, but may be feasible if AI were integrated.

Future research should also evaluate the individual features of the eHealth intervention separately to determine their potential effects on sedentary behaviour. For a larger evaluative study, such as an RCT, it would also be valuable to examine changes in ST from a 24-hour perspective to examine how time spent in sedentary behaviour is reallocated to other behaviours throughout a 24-hour day.

As the studies included in this thesis involve a limited number of participants, further research is needed so that the findings can be generalised to a larger population.

Finally, in this thesis total ST was targeted. However, given the inclusion of MCII, GAUdiS may be more effective in reducing ST which users are autonomously motivated to change. Therefore, targeting sedentary behaviour that individuals themselves perceive as unwanted or unhealthy, rather than total daily ST, may represent a way to improve intervention effectiveness. Such a focus could be viewed as consistent with the underlying values of health promotion, as it may support greater self-determination and active involvement, while also potentially yielding greater health benefits by targeting behaviours that participants themselves experience as unwanted and unhealthy.

Svensk sammanfattning

Globalt genomgår samhället en demografisk förändring som kännetecknas av en åldrande befolkning och ökad medellivslängd. Detta medför utmaningar för hälso- och välfärdssystemet. Dessa utmaningar kan mötas med hälsofrämjande interventioner för att stärka människor i att aktivt påverka sin hälsa genom hälsosamma levnadsvanor. Övergången från arbetsliv till pension innebär förändringar i levnadsvanor och kan utgöra ett potentiellt fönster för att införa hälsofrämjande interventioner. Stillasittande beteende är vanligt bland äldre och ökar risken att drabbas av flera livsstilsrelaterade sjukdomar. Forskning tyder på att eHälsointerventioner som inkluderar självhanteringsstrategier kan bidra till att minska stillasittande, men det saknas forskning och interventioner som specifikt riktar sig mot minskat stillasittande under övergången till pension. Vid utveckling av nya eHälsointerventioner kan inkludering av slutanvändare genom deltagarbaserad design bidra till att interventionen utformas i linje med användarnas behov.

Det övergripande syftet med denna avhandling var därför att bidra till hälsofrämjande arbete, genom att generera ny kunskap om hur stillasittande kan minskas och följsamhet till minskat stillasittande kan främjas genom en e-hälsointervention innehållande självhanteringsstrategier, hos äldre personer i övergången från arbetsliv till pension. Ett ytterligare syfte var att utveckla och utvärdera en sådan e-hälsointervention.

Denna avhandling omfattar fyra studier som följde en deltagarbaserad ansats. I den inledande behovsanalysfasen genomfördes fokusgruppsintervjuer med äldre personer i övergången till pension för att utforska deras uppfattningar om självhanteringsstrategier för att minska stillasittande och öka följsamheten till minskat stillasittande (Studie I). I idégenereringsfasen genomfördes workshoppar för att utforska önskade funktioner i en e-hälsointervention som inkluderar självhanteringsstrategier för att minska stillasittande och främja följsamhet till minskat stillasittande inom målgruppen. Vidare utvecklades en digital prototyp av en sådan e-hälsointervention (Studie II). I testfasen undersöktes genomförbarhet och användbarhet av den utvecklade e-hälsointerventionen med en mixed-methods single-case-studie som tillämpade en ABA-design (Studie III). I utvärderingsfasen (Studie IV) undersöktes e-hälsointerventionens effekt på stillasittande, mätt med accelerometer, i en RCT.

Resultaten belyser att deltagarna lyfte vikten av att stödja användarna autonomi i e-hälsointerventioner för att minska stillasittande. Detta genom anpassningsbara självhanteringsstrategier och att användaren själv styr över hur och när interventionen ska användas (Studier I–II). Deltagarna uttryckte att för att minska stillasittande krävs stöd för förändringar i vardagen. Självhanteringsstrategier som adresserar både affektiva och kognitiva determinanter – särskilt att hitta glädjegivande aktiviteter – identifierades som centrala för att minska stillasittande och stödja följsamhet (Studier I–II). Den utvecklade e-hälsointerventionen GAUdiS (Glädjegivande Aktiviteter Underlättar Minskat Stillasittande) innefattar funktioner för att hitta glädjegivande aktiviteter och stöd för målsättning och planering baserat på Mental Contrasting with Implementation Intensions (MCII) och kan förstås utifrån modellen Integrated Behavior Change (IBC-modellen) (Studie II). Trots att GAUdiS utvecklades tillsammans med äldre vuxna i övergången till pension som aktiva deltagare, uppvisade interventionen endast måttlig genomförbarhet och användbarhet, vilket delvis kan förklaras av tekniska problem samt de krav på tydlighet och enkelhet som ställs på GAUdiS utifrån att den är en fristående e-hälsointervention (Studie III). Den avslutande RCT-studien visade att tillgång till GAUdiS inte minskade stillasittande bland äldre personer i övergången till pension. Resultaten bör tolkas utifrån det begränsade deltagarantalet (Studie IV).

Sammanfattningsvis tyder resultaten på att stöttning av autonomi, samt att fokusera på både affektiva och kognitiva faktorer upplevs som centrala för att minska stillasittande i övergången till pension. Ytterligare komponenter utöver dem som erbjuds i den nuvarande fristående versionen av GAUdiS kan behövas för att e-hälsointervention ska vara effektiv i att minska stillasittande i övergången till pension. Givet att studierna i denna avhandling inkluderade ett begränsat antal deltagare behövs vidare forskning i större grupper för att undersöka hur stillasittande kan minskas under övergången till pension. Denna avhandling fokuserade på att minska all daglig total tid i stillasittande men framtida forskning föreslås undersöka om GAUdiS skulle kunna vara effektiv när fokus riktas mot stillasittande beteenden som individer själva upplever som oönskade eller ohälsosamma, i linje med hälsofrämjande värderingar.

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